HeforShe: Bargaining Power, Parental Beliefs, and Parental Investments*

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Abstract

We estimate a structural model of intra-household allocation of parental investments in children. We combine novel measures of verbal investments and beliefs about the returns to investment for both mothers and fathers with hyper-local variation in female bargaining power stemming from ancestral inheritance practices. Female bargaining power influences the quantity and quality of paternal investments. Specifically, fathers speak more to children, and in a more engaging way, when women have more bargaining power, but only when their wife expects investments to improve child language. Our results single out the role of economic power in mothers' ability to elicit domain-specific paternal investment. Policy simulations show large increases in paternal investments from a female empowerment policy.

Keywords: Parental investments, Child-directed speech, Female bargaining power, Parental beliefs.

JEL Codes: D13, O15, Z13

***HeforShe* is the United Nations Global Solidarity Movement For Gender Equality. "It is an invitation for men and people of all genders to stand in solidarity with women to create a bold, visible and united force for gender equality. The men of *HeForShe* aren't on the sidelines. They're working with women and with each other to build businesses, raise families, and give back to their communities." (https://www.heforshe.org/en).

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1 Introduction

Parental investments during early childhood constitute a major determinant of child human capital development (see e.g. List et al. (2018), Attanasio et al. (2022), and García and Heckman (2023) for overviews of the literature). In particular, a rich language environment in the early years is essential for young children's vocabulary and language acquisition, as well as their long-term development (Rowe, 2012; Suskind, 2015; Gilkerson et al., 2018; Bergelson et al., 2023), spurring recent interventions to foster parental verbal investments (Cunha et al., 2024; Dupas et al., forthcoming). However, the existing literature primarily focuses on *maternal* investments, overlooking both the influence of *paternal* investments and the intra-household allocation of these investments. Worldwide, over 93% of all rigorously-evaluated early parenting interventions have focused on supporting mothers exclusively (Jeong et al., 2021), even though over 70% of households with children have a father present (United Nations, 2022).

This paper studies the intra-household allocation of investments in children by *both* parents. Our analysis rests upon a unique trifecta: (i) frontier measures of mother's and father's investments, (ii) data on beliefs held by both parents about the returns and costs of these investments, and (iii) external variation in women's bargaining power stemming from deeply rooted cultural factors.

A key innovation is our measure of parental investments – speech. We measure maternal and paternal speech from child-centered day-long recordings among children aged 6 months to 4 years (26 months, on average). Day-long recordings were collected using a small device fitted to the child's clothing and then analyzed with a Voice Type Classifier (VTC, Lavechin et al. (2021)), an end-to-end neural network. This approach allows us to capture three important dimensions of investment by each parent: the overall quantity of speech, the quantity of child-directed speech, and the quality of child-directed speech, as proxied by the pitch of the parent's voice while talking to the child.[1]

Motivated by the evidence on the role of beliefs about the technology of child development in shaping parental investments (e.g., Cunha et al. (2013, 2020); Boneva and Rauh (2018); Attanasio et al. (2019); List et al. (2021); Bhalotra et al. (2024)), we elicit both parents' beliefs about the production function of language development, as well as their perceived costs of talking to children.[2] In addition, our field setting, the Solomon Islands, offers extremely rich variation in traditional cultural practices even among otherwise ethno-linguistically ho-

[1] Emotional expressiveness, particularly higher pitch, is key in maintaining a child's attention and fostering social and cognitive engagement (see e.g., Saint-Georges et al. (2013); Ramírez et al. (2020)).

[2] We are only aware of the work of Almås et al. (2024) who elicit beliefs from fathers as well as mothers about the importance of certain parental inputs in Tanzania.

mogeneous communities (BenYishay et al., 2017). This enables us to leverage hyper-local variation in female bargaining power based on matrilineal inheritance, in which daughters inherit land from their mothers. We provide evidence that matrilineal inheritance is a reliable proxy of female bargaining power, with women enjoying more agency and leisure, even among otherwise similar villages.[3] We also rely on other cultural determinants of female bargaining power, such as bride price (see, e.g., Anderson (2007); Ashraf et al. (2020); Lowes and Nunn (2018)).

We begin our analysis by modeling intra-household allocation of parental speech investments. We assume a collective household model in which the mother's (Pareto) weight in the collective utility function is higher under a traditionally matrilineal inheritance structure. Both parents derive utility from child language development but bear a cost associated with speech investments. Parents have imperfect information about the language production function and form individual-specific beliefs about the return of their investment. The model provides insights into how expected returns, perceived cost of investment, and female bargaining power influence paternal investments in children. Specifically, it predicts that both the father's investment and the elasticity of his investment with respect to the mother's expected returns increase with the mother's Pareto weight. This latter prediction, which captures the subtle interplay between beliefs and bargaining dynamics beyond the simple effect of inheritance practices, is the main focus of our analysis. We structurally estimate the parameters of the model to test its predictions and perform counterfactual analysis.

Empirically, we document large gaps in investments between parents. Mothers talk twice as much to children and elevate pitch more when talking to children, indicating more intense engagement during communication. Consistent with the model's predictions, fathers in matrilineal villages speak more than fathers in patrilineal villages. We also find that both parents report positive returns, of similar magnitudes, and positive costs of investments. Yet, the findings reveal distinct pathways for paternal and maternal investments. Fathers are more responsive to their perceived costs of investment, while mothers respond primarily to their expected returns.

Our main result is that women who expect a higher return are able to elicit more paternal speech investment when they have greater bargaining power: the elasticity of paternal investments to mother's expected returns increases in female bargaining power. A one standard deviation increase in mothers' beliefs about the returns to parental speech increases fathers' speech investments by 0.55 standard deviations in matrilineal villages. This result holds for several measures of speech input and for quality, and is consistent with the ethnographic

[3]See also, e.g La Ferrara (2007); Gneezy et al. (2009) and Lowes (2022) for evidence on the connection between matrilineal inheritance and female status.

record showing that women in societies with matrilineal inheritance have greater leverage to demand higher paternal investment (see e.g. Hrdy (2016, 2024)).

Turning to mechanisms, we find that while husbands of more powerful women do spend more *time* taking care of children, they only invest more in *speech* when mothers have high expected returns to child-directed speech. This illustrates how the quantity and quality of paternal investments are specific to the domains where mothers expect high returns. Further, we show that the additional paternal investment is primarily directed towards girls, in line with maternal preferences over child gender. We rule out several other channels that could explain the relationship between maternal beliefs and father speech in matrilineal villages, including assortative matching and learning. In particular, we show that husbands of high-returns mothers in matrilineal villages are observationally similar to others, including in terms of beliefs and cost of speech, education, income sources, and social preferences. High-returns mothers in matrilineal villages, as well as their children, are also similar to others. Finally, we pinpoint the role of *economic* power in driving paternal investment as opposed to other potential dimensions of female status, including kin and family networks due to residence patterns (Khalil and Mookerjee, 2019; Bau, 2021). We provide additional evidence on the influence of *individual* economic power by leveraging quasi-random variation in the death of the mother's parents and within-village variation in bride price.

We discuss the quality of our probabilistic beliefs data and analyze the sensitivity of our results to specification choices. We show that our results are robust to different assumptions about the functional form of the relationship between Pareto weights and matriliny, as well as different assumptions about the cost function. In particular, our results are insensitive to the set of conditioning variables at the child-, parental-, household- or village-level that enter the cost function, as well as considering altruistic individuals who internalize their spouse's cost in their utility function. Our results are also robust to different assumptions about the structure of the error terms.

Our paper contributes to the economics of parenting. Parents play a crucial role in shaping children's early development (e.g., List et al. (2018)), and there is a growing body of literature examining the determinants of parental investments in children (see Cunha and Heckman (2007) for an early contribution and Doepke et al. (2019) and Doepke and Zilibotti (2019) for recent reviews). However, the literature on parenting offers only limited evidence on fathers' investments. Most theoretical models only consider a single parent and a single child (see discussion in Doepke et al. (2019)), and empirical studies predominantly examine mothers.[4] Our contribution is to analyze investments made by both mothers and fathers, as

[4]See Evans and Jakiela (2024) for a recent survey of the emerging literature on fathers' involvement in low and middle-income countries, and Doepke and Zilibotti (2019) for data on parenting styles from both

well as the dynamic of intra-household allocation of these investments.

We achieve this through a novel approach, combining a frontier and objective measure of investment that is consistent across parents, with precise data on both parents' beliefs regarding investment returns and the associated costs. A basic identification problem in understanding the drivers of parental investment is that observed investments are consistent with many alternative specifications of preferences, expected returns, perceived costs, and bargaining power (Manski, 2004). Our data allow us to disentangle their respective roles and shed new light on how female bargaining power affects paternal investments. This data collection effort answers recent calls by e.g., Almås et al. (2024) and Caplin (forthcoming) for the measurement of typically unobserved constructs in order to improve inference on decision-making without overly restrictive assumptions.

This paper also makes several contributions to the literature on the importance of social structures for the status of women and economic development (e.g. Greif (1994, 2006); Alesina and Giuliano (2014); Henrich (2016); Bahrami-Rad et al. (2022)). While most previous evidence highlights how social structures affect group-level outcomes, our approach and data microfound such aggregate relationships. We provide the first application of a structural model to understand how deeply rooted cultural variables affect decision making. We model social structures as a bargaining power shifter and we exploit rich variation in traditional inheritance, residence, and marriage patterns that teases out various dimensions of female status and isolates the role of economic bargaining power. Related literature connects female-biased cultural practices, such as matrilineal inheritance (La Ferrara, 2007; Gneezy et al., 2009; La Ferrara and Milazzo, 2017; Lowes, 2021, 2022), matrilocal residence (Bau, 2021; Khalil and Mookerjee, 2019), and bride price (Anderson, 2007; Ashraf et al., 2020; Lowes and Nunn, 2018) to female bargaining power, female autonomy and status, as well as parental investments in children. However, this literature typically leverages variation at the ethnicity level. We instead rely on within-ethnicity variation, attenuating potential concerns that variation in specific practices confounds other cultural or institutional differences across ethnic groups.

Our focus on female bargaining power contributes to the literature on women's empowerment and its implications for intra-household decisions toward children. Earlier studies indicate that female bargaining power is associated with higher spending and better nutrition for children (e.g., Thomas (1990); Lundberg et al. (1997); Duflo (2003)). We show how female bargaining power also shapes the allocation of non-financial, time-intensive investments in children.

mothers and fathers in high-income countries. See also Giannola (2024) for an example of how parents allocate resources between siblings.

Our findings have several implications for the design of policy interventions. Early childhood development interventions can have substantial impacts on children's cognitive and socioemotional development, but these interventions so far overwhelmingly rest on mothers. Given that time allocation over household activities and child rearing is already a major dimension of gender inequality even in the most advanced economies (Kuziemko et al., 2018), these interventions have recently spurred concerns about exacerbating gender inequalities by further restricting female labor force participation and imposing additional childcare commitments on mothers (Evans et al., 2021). Yet, so far, with only scant evidence on fathers' investments and decision-making, involving fathers in early childhood programs has proven challenging (Garcia et al., 2022). We begin to fill this gap in the literature.

Our policy simulations suggest that female empowerment policies are the most effective at increasing paternal investments in child development. For instance, an intervention equivalent to transitioning from fully patrilineal to equal inheritance leads to a 17% increase in fathers' verbal engagement with their children, significantly outpacing the impact of alternative interventions. Additionally, reducing fathers' costs of engaging with their children—such as through playgroups that teach enjoyable activities—modestly increases fathers' vocalizations (by 1–2%). By contrast, information interventions targeting fathers' beliefs about the benefits of verbal engagement yield no measurable effects. However, informing mothers about these benefits not only increases their own verbal engagement but also generates positive spillovers to fathers. Combining increasing women empowerment and expected returns with reducing father's cost leads to a 21–22% increase in paternal vocalizations in patrilineal settings and 6–8% in matrilineal ones. Our analysis and results, overall, highlight the importance of measuring potential policy spillovers across parents.[5]

2 Modeling parental speech investments

We present a simple model of intra-household allocation of parental investment showing the role of bargaining power, parental expectations about child development, and parental cost of investment. We build on the seminal collective model of Chiappori (1988), which starts from the basic assumption that in a multi-person household, household members have their own utility functions but cooperate in a bargaining process that results in a Pareto-efficient allocation.

We consider a household with a mother m, a father f and one child c. Each parent i derives utility from the child's human capital, proxied by language development Θ, and

[5]See Guo et al. (2024) for evidence of spillover effects from an early childhood development program, where untreated children benefited through their parents' social connections to families receiving the intervention.

incurs a cost represented by a twice-differentiable convex cost function $C_i(x_i)$ associated with their time-intensive investment x_i. Parent i's utility is given by:

$$U_i(x_i, \Theta) = \omega_i \Theta - C_i(x_i), \quad i = m, f \tag{1}$$

where ω_i is parent i's utility associated with the child's language development.

Looking at the household decision-making, we assume a collective model (Chiappori, 1988) in which the household utility function is a weighted sum of the mother's and father's expected utility function, U_m and U_f respectively:

$$U = \alpha U_m + (1 - \alpha) U_f \tag{2}$$

The parameter α, often referred to as a Pareto weight, captures the relative importance given to the mother's utility in the couple's utility function. In our context, α is assumed to vary with the inheritance structure in the village where the couple lives. The key characteristic of the collective model is that household decisions are efficient. This efficiency is achieved by maximizing the household utility function U subject to household constraints. As standard in these models, we assume common knowledge of individual preferences and expectations.

The household faces a language production function constraint.[6] The objective production function of child c's language is as follows:

$$\Theta(x, \theta_0) = \mu_0 + \mu_1 \theta_0 + \mu_2 x + \zeta_c, \tag{3}$$

where θ_0 is the child's human capital endowment at birth, x is the total investment received by the child,[7] and ζ_c denote a mean-zero variable unknown to the parents at the time the investment decision is made. The parameter μ_0 is a constant term, the parameter μ_1 captures the productivity of initial endowment, the parameter μ_2 measures the elasticity of the investment in producing language development, while ζ_c are unexpected shocks that influence child development, such as the onset of an illness.[8]

The standard model assumes that parents know the actual production function. We relax this assumption and allow each parent to have their own subjective expectations over the

[6] A branch of the literature seeks to identify and estimate the actual production function from data on child development. Our purpose here is instead on understanding the drivers of intra-household allocation of investment.

[7] The assumption that the production function depends on total investment supposes that father's and mother's speech inputs are perfect substitutes. However, as we discuss below, the assumption of perfect substitutability is not key to our results.

[8] Existing work typically assumes linearity, CES or Cobb-Douglas production function (e.g., Cunha et al. (2013); Tincani et al. (2022)).

parameters describing returns to investment. The production function that parent i *perceives* is given by:

$$\Theta_i(x, \theta_0) = \eta_{i,0} + \eta_{i,1}\theta_0 + \eta_{i,2}x + \xi_i, \quad i = m, f, \tag{4}$$

where $\eta_{i,j}$ are individual-specific beliefs about the production function parameters, x is the total investment received by the child from both parents, and ξ_i is a zero-mean variable that captures beliefs uncertainty and is independent of investments.

Given these beliefs, parent i's expected utility is based on their own perceived production function, formulated as:

$$\mathcal{E}U_i((x_i, \Theta_i(x_f + x_m)) = \omega_i(\eta_{i,0} + \eta_{i,1}\theta_0 + \eta_{i,2}(x_f + x_m)) - C_i(x_i), \quad i = m, f. \tag{5}$$

The couple will choose the investment levels x_f and x_m to maximise the household expected utility subject to two effort constraints $x_i \leq X_i, i = m, f$. The expected household utility function is given by:

$$\begin{aligned}\mathcal{E}U = \alpha\big[\omega_m\big(\eta_{m,0} + \eta_{m,1}\theta_0 + \eta_{m,2}(x_m + x_f)\big) - C_m(x_m)\big] \\ + (1-\alpha)\big[\omega_f\big(\eta_{f,0} + \eta_{f,1}\theta_0 + \eta_{f,2}(x_m + x_f)\big) - C_f(x_f)\big].\end{aligned}$$

Assuming an interior solution, the first-order-conditions with respect to x_m and x_f are given by:

$$\frac{\partial \mathcal{E}U}{\partial x_m} = \alpha\left(\omega_m\eta_{m,2} - C_m'(x_m)\right) + (1-\alpha)\omega_f\eta_{f,2} = 0. \tag{6}$$

$$\frac{\partial \mathcal{E}U}{\partial x_f} = \alpha\omega_m\eta_{m,2} + (1-\alpha)\left(\omega_f\eta_{f,2} - C_f'(x_f)\right) = 0. \tag{7}$$

Our model first establishes that the optimal investment strategy of a parent is increasing in both their and their spouse's expected returns $\eta_{m,2}$ and $\eta_{f,2}$ and decreasing in their own marginal cost (see equations (6) and (7)). The FOCs also show that the father investment is increasing in the mother's Pareto weight α (and vice versa for mothers).

We are also interested in how the father's investment x_f responds to changes in the mother's belief parameter $\eta_{m,2}$. We use implicit differentiation on the FOC for x_f with respect to $\eta_{m,2}$:

$$\frac{\partial}{\partial \eta_{m,2}}\left(\alpha\omega_m\eta_{m,2} + (1-\alpha)\left(\omega_f\eta_{f,2} - C_f'(x_f)\right)\right) = 0. \tag{8}$$

Solving for $\frac{\partial x_f}{\partial \eta_{m,2}}$, we have:

$$\frac{\partial x_f}{\partial \eta_{m,2}} = \frac{\alpha \omega_m}{(1-\alpha) C_f''(x_f)}. \tag{9}$$

Thus, the elasticity of the father's investment to the mother's beliefs $\frac{\partial x_f}{\partial \eta_{m,2}}$ is increasing in her pareto weight α (and vice versa for mothers). This means that paternal investment will be more responsive to the mother's expected return when she has higher bargaining power.

In Section 4, we estimate the structural parameters of the model, leveraging unique data on parental investments x_f and x_m, parental beliefs $\eta_{f,2}$ and $\eta_{m,2}$ and shifter of the Pareto weight α.

The model is simple, designed to profile parents' decision-making process over their respective time-intensive investment in children and the role of bargaining power and beliefs. The simplifying assumption that parental investments are perfect substitutes is not key to our results, as long as the beliefs about the elasticities of child language development to own and spouse's investments are positively correlated. Our model omits other potentially important aspects, such as multidimensional skills, the role of financial inputs, the child's agency, and labor force participation. These simplifications are suited to our context. As we discuss in more detail in the next section, our focus is on early childhood (6 to 48 months), where the child has yet no agency and speech development is the primordial skill, and in a subsistence economy context, where opportunities for financial investments in children and formal employment are limited.

3 Background and Data

Our study site is located in the Solomon Islands, in the South Pacific, which comprises six major islands and over 900 smaller ones. The people of the Solomon Islands are Melanesian, and display great cultural diversity, with 72 distinct living languages (according to Ethnologue) for a total population of around 800,000 people. The Solomon Islands ranks 156th out of 193 countries on the Human Development Index, reflecting ongoing challenges in health, education, and living standards (United Nations Development Programme (2024)).

3.1 Sample

We conducted our data collection from June to August 2019 in 44 villages across two provinces, Western and Choiseul.[9]

The villages in our sample are small, isolated, coastal lowland communities that depend mainly on subsistence horticulture and fishing for their livelihoods. Only 4.85% of respondents in our sample derive any income from paid work. On average, people in our sample work in their gardens a couple of times a week and sell goods in the market less than once a month. In addition, men fish a couple of times a week and women a few times a month. An average village consists of 87 households and 441 people, most of whom have no access to grid electricity, running water, or improved sanitation. Descriptive statistics in Table A1 show that electricity is "used by most households" in only 7% of villages and only 3% of households have access to running water.[10] Only half of the households have a water tank, and only 25% of the villages always have enough drinking water for all households. The main mode of transport is by ship or outboard canoe and, in general, access to roads is extremely limited. The average travel time between the villages in the sample and the provincial capital is six and half hours, and the average travel time to the country's capital city Honiara, on the island of Guadalcanal, is two and a half days, leaving most communities isolated from major market centers.

Through public lotteries in each village, we randomly selected a total of 820 adults. A subset of 182 married couples with at least one child aged 6 months to 4 years old, for whom we surveyed both the wife and the husband, constitutes our analytical sample. In a few cases in which the parents had more than one child within the target age range, we collected information on the additional children, for a total of 196 children.[11] The average age of the children in our sample is 26 months, the median 24 months, and 98.15% of children are less than 3 years old. Mothers are 29 years old on average, and fathers, 33 years old. Appendix E provides further details on the sample and survey data. We conducted individual surveys with both parents, including a dictator game and competition task, and recorded the target child over the course of an entire day.

We collected day-long child-centered recordings using a wearable device, an increasingly used technique in early language development research (Lavechin et al., 2021). In our study,

[9] There were two survey teams in Western and one in Choiseul, each consisting of two research assistants and two local enumerators, with at least one female and one male, who could speak English, Pijin (the lingua franca), and at least one local language. The composition of the teams did not change over the course of the data collection.

[10] We take access to a flush toilet as our proxy of access to running water. 75% of households use the bush, the river or the sea as their "main type of toilet".

[11] We check that our results are invariant to dropping the 14 additional children, or to clustering standard errors at the household level to account for correlated observations across siblings.

each child wore a t-shirt fitted with two small breast pockets, into which a pair of USB voice recorders were inserted. Two recorders were used because some devices may stop functioning in field conditions with high humidity and variable temperatures. We provide more information on the recording technology in Appendix Section C. Parents were informed that the study aimed to explore how children acquire language and assured that the content of their conversations would not be listened to. We obtained 377 recordings for 196 children. Compliance was very high, with only one family not consenting to the recording, and recordings lasting 15.4 hours on average (SD 4.3 hours, range 0.11 minutes - 24 hours). Less than 5% of the recordings lasted fewer than 6 hours, and this is similar by matriliny (see Appendix Table A2).

We focus on early childhood as a critical time for language acquisition. Young children are particularly sensitive to language stimulation, the investment we focus on (Dickinson and McCabe, 1991). Any shortfall at this stage can have lasting effects due to the brain's remarkable plasticity. Differences in early conditions often become self-reinforcing due to dynamic complementarity with subsequent investments (Cunha and Heckman, 2007). In addition, our focus on early childhood facilitates a cleaner analysis by limiting child agency and the type of parenting decisions (such as parenting styles, e.g., Doepke and Zilibotti (2017)).

Language acquisition is a particularly central element of cultural transmission in the Solomon Islands, where children are expected to learn several languages from an early age. For the very young children in our sample, language acquisition is the most central skill. Beyond language diversity, the Solomon Islands also have important oral traditions that are transmitted to children and include origin stories, genealogies, songs, customary knowledge, reminiscences about the past, and spells and prayers, including many dating from the pre-Christian era.

3.2 Matrilineal inheritance in the Solomon Islands

Recent anthropological research suggests that cultural diversity was a defining feature of pre-contact small scale societies (see, e.g. Diamond (2013); Graeber and Wengrow (2021)). Anthropologist Marshall Sahlins, reflecting on the Pacific Islands, noted how "the native peoples of Pacific Islands unwittingly present to anthropologists a generous scientific gift: an extended series of experiments in cultural adaptation" (Sahlins, 1963). The Solomon Islands, in particular, provide a context of uniquely rich and hyper-local variation in land inheritance practices, even *within* ethno-linguistic cultural groups that are otherwise similar in their family structure and cultural characteristics (BenYishay et al., 2017). The persistence of

cultural diversity in the Solomon Islands owes to a large extent to the limited reach of the centralized state.

In matrilineal villages of the Solomon Islands, daughters inherit land from their mothers. This form of matrilineal land inheritance is the norm in matrilineal societies of the Pacific – with genetic, linguistic and archaeological evidence establishing the ancestral prevalence of matriliny dating as far back as 5,000-4,500 BP (Jordan et al., 2009) – as well as in some societies of southern and central Africa and some native American cultures.[12] Today, nearly 90% of the land is still owned customarily, vested either in matrilineal or patrilineal clans and tribes (Stege, 2008; United Nations Development Programme (UNDP), 2021). Matrilineal inheritance is combined with significant responsibility of women over land (Stege, 2008) and substantial female empowerment.[13]

A given inheritance system is dominant within a village but may coexist with another system. Accordingly, we measure matrilineal bias as a categorical variable, M_h, which takes a value of 0 in exclusively patrilineal villages (41.5% of our sampled villages), 0.25 in villages where patrilineal inheritance is dominant but not exclusive (26.8%), 0.50 in villages where matrilineal and patrilineal inheritance are equally prevalent (7.3%), 0.75 in villages where matrilineal inheritance is dominant but not exclusive (14.6%) and 1 in exclusively matrilineal villages (9.8%).[14] The mean value of M_h is 0.30, with a standard deviation of 0.36.

In our sample, women in matrilineal villages are more likely to have inherited land (Panel A of Table 1) and, consistent with the literature on the positive consequences of matriliny for female bargaining power and outcomes (Lowes, 2021, 2022), they enjoy more leisure time and have later ages at first marriage (by 1.7 years). Women in matrilineal villages keep 10 percentage points more of the dictator game endowment when they play with their spouse, compared to women in patrilineal villages, an indication of a greater economic power within the household (Gnagey et al. (2018)).[15] We also observe that the distribution of household assets is tilted towards female preferences in matrilineal villages. Households in matrilineal

[12] In other matrilineal cultures, such as the ones studied by La Ferrara (2007), land is transferred from the mother's brother to his sister's son. This is prevalent in areas that practice avunculocal residence, where a male child is expected to join his maternal uncle's residence. Avunculocal residence is a minority practice in matrilineal systems and is absent in our sample. Consequently, in our sample, uncles are hardly involved in their sisters' children's lives, even less so than unrelated individuals, with no difference by matriliny (Cassar et al., 2022).

[13] As stated by a woman of nearby Palau island, "the women here are so sure of themselves ... maybe it's that we know for sure that we have land ... Even if I don't get land from my husband, I still have it from my mother and nothing can change that." (cited in Stege (2008)).

[14] The matrilineal bias variable codes the village leader's answers to the question: "How is land inherited in this village?" - (i) father; (ii) mother; (iii) both, but mostly father; (iv) both, but mostly mother; (v) both, mother and father equally; (vi) other. We check the robustness of our results to coding M_h as a categorical variable. See in particular Figure 2.

[15] We provide more detail on all survey and experimental measures in Section E.

villages are more likely to own a water tank and a toilet, but there is no difference in ownership of a mobile phone, boat, radio, or solar lamp.

Importantly, our data confirms that matrilineal and patrilineal villages are otherwise comparable in cultural and political organization (Table 1). We observe no differences in the participation of women vs. men in formal village governance structures nor generalized trust levels. We also observe no difference in spouses' tendency to compete with one another, a measure of spousal cooperation (Lowes, 2022), suggesting that household harmony does not vary with matriliny in our context. There is also no statistically significant relationship between fertility and matrilineal inheritance.[16]

3.3 Expected returns and costs of child-directed speech

We measure both mothers' and fathers' beliefs about the returns to speaking to children with respect to children's outcomes. Specifically, we elicit the subjective probabilities associated with a child achieving a language development milestone in two hypothetical scenarios: one where the caregiver engages in frequent communication (high investment) and another where communication is infrequent (low investment). We employ a visual aid approach commonly used in developing countries. Respondents are provided with 10 beans to express their beliefs in a probabilistic format, where each bean represents one chance out of 10 (Delavande, 2023). We adopt the language milestone utilized by Bhalotra et al. (2024), specifically, the expectation that a child can string together 2-3 words by the age of 2. Data Appendix E details the exact wording and Appendix B.1 discusses the data in more detail.

Appendix Figure A1 shows the distribution of parental beliefs for high and low investments. There is considerable heterogeneity in beliefs. Overall, parents expect a higher probability of a child speaking 2-3 words by age 2 under the high investment scenario. The mean answer is 7 for both mothers and fathers in the high-investment scenario and 3 in the low-investment scenario. In the US context, 72% of children speak partial sentences by age 2 (Cunha et al., 2020), suggesting that our respondents' average expectations are well calibrated. We present some data quality checks in Appendix B.1

We transform the data into expected returns, taking the difference in expected outcomes between the high and low investment cases at the individual level.[17] Men and women expect, on average, that frequently talking to the child, as opposed to rarely, will increase the

[16] In our sample, when considering the total number of children of women aged 18 to 45 (inclusive) as an outcome, controlling for age and province fixed effects, the point estimate associated with M_h is -0.108 with a standard error corrected for clustering at village level of 0.256 (P-value: 0.673).

[17] If parental investments are perceived as perfect substitutes, the expected return is a measure of $\eta_{i,2}$ in 5. If parental investments are perceived as imperfect substitutes, the expected return is a weighted average of the subjective elasticities of father's and mother's investments in the child language production function.

likelihood of the child speaking sentences by age 2 by 43 percentage points. The distribution of expected returns for men and women is, on average, similar (p-value of 0.483 on a Kolmogorov-Smirnov test of equality of distributions shown in panel (a) of Figure 1).

We also examine the distribution of parental expected returns *within* a couple. This is important because, if parental beliefs were perfectly correlated with one another, we would be unable to measure their respective influences on parental investment or the influence of spousal bargaining power. Correlation of parental returns could either result from assortative matching on beliefs, spousal communication and persuasion, or learning from the same environment (including from their child's specific ability). Our data reveals that spouses have distinct beliefs about returns, with a raw correlation of -0.004. Panel (c) of Figure 1 shows the complete distribution of couples according to each partner's return. Very few couples are along the 45-degree line, where the distribution would fall if beliefs were perfectly correlated. There is no extra mass in the top right quadrant (high returns mothers matched with high returns fathers) nor in the bottom left quadrant (low returns matched to one another).

We elicit mothers' and fathers' perceived cost of talking to their child on a scale from 0 to 10 (see Appendix E). Perceived costs are slightly higher for mothers compared to fathers (p-value of 0.005 on a Kolmogorov-Smirnov test of equality of distributions, see Panel (b) of Figure 1). Spousal correlation in costs of speech is slightly higher than for returns, but still low (0.209) (see also the distribution of spousal costs in Panel (d) in Figure 1).

Consistent with the absence of other cultural and institutional differences, parental beliefs about returns and costs are similar across matrilineal and patrilineal villages, as shown in Table B1. We also observe no differences by matriliny in the absolute gaps between mothers' and fathers' expected returns and costs (Table A3). A small absolute gap suggests assortative matching on beliefs and costs, or effective spousal communication concerning these factors. Thus, our findings indicate no variation in assortative matching or spousal communication based on inheritance patterns.

3.4 Speech as a measure of parental investments

Speech recordings were analyzed with a Voice Type Classifier (VTC, Lavechin et al. (2020)), an end-to-end neural network trained with a combination of various child-centered corpora of children aged 0-4 years in multilingual environments, including languages spoken in the Pacific, which match the conditions of our study.[18] A precursor of this method, called LENA, has been used for over a decade in the USA (for a meta-analysis see Wang et al., 2020) and in a recent study of an information intervention targeting mothers in urban Ghana

[18] Twelve different languages are spoken across our sample. Children grow up in multilingual environments, including at least one local language and the lingua franca.

(Dupas et al., forthcoming). Our method relies on more recent neural network architectures, which allow a more accurate classification of female and male voices (F-score of 43% versus 37%; Lavechin et al., 2020).[19] In Appendix Section C, we discuss how VTC compares with alternative technologies, including manual annotations.

For every 10 ms frame, VTC returns whether the key child (i.e., the child wearing the recording device), another child, an adult female, or an adult male is vocalizing. A vocalization is then defined as a sequence of frames where the relevant individual vocalizes. Following standard practice in the field (Cristia et al., 2020), we counted the number of vocalizations attributed to the relevant speaker over the whole recording. We then divided that by the length of the recording and re-scaled to 12 hours to produce a per day figure that accounts for 12 hours of sleep.

Our main outcomes of interest to measure parental investments are the resulting normalized adult female and male **vocalizations per day**.[20] These vocalizations may include both speech directed at the child and speech overheard by the child. Yet, existing literature has shown that the quantity of adult vocalizations, regardless of whether they are directed at the child or not, is positively correlated with children's production of speech-like vocalizations, suggesting that overall adult speech is a relevant input into child language development (Bergelson et al., 2023).[21]

Additionally, recognizing the importance of child-directed speech and high-quality interactions for child development (e.g., Weisleder and Fernald (2013); Suskind (2015)), we use two supplementary measures of the *quantity* of child-directed speech: **conversational turns** and **vocalizations at high pitch**. To capture the *quality* of child-directed interactions, we use a measure of **pitch** when talking to children. Child-directed speech, often called motherese or parentese, is characterized by higher pitch, a pattern that is near-universal according to a recent systematic review (Cox et al., 2023).[22] This type of speech captures young children's attention and promotes engagement. Studies demonstrate that infants respond more positively to high pitch speech, reinforcing parental use and fostering a feedback loop that enhances communication, supporting early language development, turn-taking in conversa-

[19] F-score is a measure of accuracy that goes from 0 to 100%, higher is better. See exact definition in Appendix D.

[20] A potential concern with this measure is that it might reflect proximity to the child rather than actual stimulation. In our exploration of mechanisms, we show that our results are not mediated by how much time the father spends with the child.

[21] In a study involving children aged 2 to 48 months from 12 countries across six continents and varied living contexts, Bergelson et al. (2023) find that for every 100 adult vocalizations per hour, children produced 27 additional vocalizations. In our data, children produce 52.9 (30.5) more vocalizations per day for every 100 female (male) vocalizations per day. See Appendix Table A4.

[22] To our knowledge, the only language in which a higher pitch for child-directed over adult-directed speech has not been observed is Quiche' Mayan (Pye, 1986).

tions, and vocabulary growth (e.g., Saint-Georges et al. (2013); Ramírez et al. (2020)). Pitch is also a strong predictor of the emotional connection between caregiver and child (Smith and Trainor, 2008).[23]

The first measure of child-directed speech, Conversational Turn Count or CTC, is commonly used in studies employing LENA (see e.g. Romeo et al. (2018), Dupas et al. (forthcoming) and Cunha et al. (2024)). CTC tallies child-adult or adult-child vocalization sequences, aiming to capture conversation. However, CTC may be a noisy measure, as it may count instances that are not actual conversations, such as a child vocalizing after an adult speaks, regardless of whether the child is responding, interrupting, or simply talking to themselves, a likely event given the young age of children in our sample (6 to 48 months).[24]

To improve on the CTC measure and increase the likelihood that we capture a child-parent interaction, we analyze the pitch characteristics of adult vocalizations. We measure the median pitch of each vocalization within a recording and then calculate the average of these median pitches for each individual. Our analysis uses on two pitch-related measures: the total number of vocalizations that exceed the average pitch, as a measure of *quantity* of child-directed speech; and the average median pitch during conversational turns, as a measure of *quality* of child-directed speech.[25]

The current state of the technology does not make it possible to distinguish further between female or male voices. From hereon, we consider female (male) vocalizations as mothers' (fathers') vocalizations. This approximation may introduce measurement error but is not unreasonable given that children in our sample spend the vast majority of their time with their parents. Other than parents, but far behind, children receive the most care from their maternal grandmother and from neighbors and friends, who on average take care of the child between once a week and several times a month in our sample (see Cassar et al. (2022) for more detail). To deal with such potential measurement error, we account for the frequency of care by female or male individuals other than the parents in some specifications.[26]

Descriptive Statistics

[23] In Appendix Table A5, we confirm that pitch is higher during a conversational turn with a child.

[24] One difference between our measure of CTC and the LENA measure of CTC is that LENA throws out overlapping speech between adult and speech, while our measure counts it as CTC. This explains why mean CTC is higher in our sample compared to, e.g., Dupas et al. (forthcoming).

[25] Pitch is therefore missing if no conversational turn occurs during the recording. This is true for 6 recordings, all of which have a duration of less than one minute.

[26] We will also show in columns 12 to 14 of Table A9 that care by people other than the parents is unrelated to mothers' expected returns, matriliny and their interaction. Potential measurement errors therefore would be orthogonal to our main variables of interest.

Table 2 presents descriptive statistics of the four measures of parental speech. The means reported at the bottom of columns 1 and 5 of the table show that mothers vocalize overall more than twice as much as fathers (2,783 vocalizations per day versus 1,312). Mothers have also notably more conversational turns than fathers (1,415 versus 442 per day) and more vocalizations at high pitch (1,163 versus 584 per day). Unsurprisingly, given sexual differences, mothers' average pitch is higher than fathers' (29 vs. 23 semitones, see columns 3 and 7). Nevertheless, as shown in columns 1 and 2 of Table A5, mothers also *elevate* pitch more than fathers during a conversational turn with children, suggesting more intense engagement with children when talking to them.[27]

We next explore how adult speech varies with matrilineal inheritance. The multilingual nature of our sample raises the potential concern that languages spoken in matrilineal villages may systematically differ in a way that could confound parental speech.[28] We show evidence arguing against this possibility in Table 2: none of our speech measures varies significantly with matriliny for women. As shown in columns 2 and 4 of Appendix Table A5, women also do not elevate pitch more during a conversational turn in matrilineal vs. patrilineal villages. Moreover, we will show that our results are robust to the inclusion of language fixed effects.

In contrast to speech by mothers, fathers have significantly more vocalizations overall, more vocalizations at high pitch, and more conversational turns in matrilineal villages (Table 2). Average male pitch during a conversational turn is not statistically significantly different across villages, indicating that once fathers engage in conversation, their level of pitch is comparable, which is also consistent with the absence of linguistic differences across villages. The magnitude of the differences in paternal speech investments is large: fathers have 27.2% more vocalizations, including 29.5% more vocalizations at high pitch, and 21.4% more conversational turns with children in matrilineal villages. We now explore why.

4 Estimation

Our objective is to estimate the optimal mother and father investments given by equations (6) and (7). To ensure tractability, we impose several parametric assumptions. First, we assume that the convex cost function is given by $C_i(x_i) = \chi_{i,1}x_i + \chi_{i,2}\frac{x_i^2}{2}, \chi_{i,1} \geq 0, \chi_{i,2} \geq 0$. Under this functional form assumption, the optimal level of investment x_m^* and x_f^* are linear

[27]The coefficient associated with the interaction between mother and CTC in a regression of median pitch of each vocalization (corresponding to the difference in the coefficient associated with CTC between columns 1 and 3 in Table A5) is positive and statistically significant at the 1% level.

[28]The Solomon Islands displays great linguistic diversity, with 12 different languages spoken across the villages in our sample.

in perceived returns and given by:

$$x_m^* = \frac{\omega_m}{\chi_{m,2}}\eta_{m,2} + (\frac{1-\alpha}{\alpha})\frac{\omega_f}{\chi_{m,2}}\eta_{f,2} - \chi_m \tag{10}$$

$$x_f^* = \frac{\omega_f}{\chi_{f,2}}\eta_{f,2} + (\frac{\alpha}{1-\alpha})\frac{\omega_m}{\chi_{f,2}}\eta_{m,2} - \chi_f \tag{11}$$

where $\chi_i = \frac{\chi_{i,1}}{\chi_{i,2}}, i = m, f$.

Second, we assume that the cost parameters χ_m and χ_f are a linear function of the individual perceived cost of talking to children (see section 3.3), as well as parent, child, and household characteristics, which we describe below, so that: $\chi_i = \kappa_{1,i}C_i + \gamma_i X_{i,c,h}, i = m, f$.

Third, we rely on matrilineal inheritance as a bargaining power shifter. In particular, we assume that $\frac{\alpha}{1-\alpha}$ is linearly related to the prevalence of matrilineal inheritance in the village. Specifically, we model this relationship as: $\frac{\alpha}{1-\alpha} = \beta M_h$, $\beta > 0$ (implying $\frac{1-\alpha}{\alpha} = \frac{1}{\beta M_h}$),[29] so that the mother's Pareto weight α is increasing in matrilineal inheritance prevalence.

Fourth, we assume there is an additive measurement error in our parental speech measures. This error is assumed to be normally distributed with a mean of zero. Since the measurements of maternal and paternal investment are derived from the same recording, we allow for the possibility that the measurement errors may be correlated between maternal and paternal investments.

We test the robustness of our estimates to alternative parametric assumptions in Appendix Section B. In particular, we allow for spousal altruism and for different functional form assumptions for the relationship between matrilineal inheritance and female bargaining power, namely relaxing the linearity assumption.

We estimate the optimal parental investment using Seemingly Unrelated Regressions (Zellner, 1962):

$$\begin{aligned} Speech_{m,c,h} = a_m + \delta_{1,m}\eta_{m,2} + \delta_{2,m} * \eta_{f,2} + \delta_{3,m}\frac{1}{M_h} * \eta_{f,2} \\ + \kappa_{1,m}C_m + \gamma_m X_{m,c,h} + \epsilon_{m,c,h} \end{aligned} \tag{12}$$

$$\begin{aligned} Speech_{f,c,h} = a_f + \delta_{1,f}\eta_{f,2} + \delta_{2,f} * \eta_{m,2} + \delta_{3,f}M_h * \eta_{m,2} \\ + \kappa_{1,f}C_f + \gamma_f X_{f,c,h} + \epsilon_{f,c,h}, \end{aligned} \tag{13}$$

with $cov(\epsilon_{m,c,h}, \epsilon_{f,c,h}) = \sigma_{m,f}$.

Each observation $Speech_{i,c,h}$ is a measure of speech by parent i to child c in household h. $\eta_{i,2}$ is i's beliefs about the return to speaking to children. M_h measures the prevalence of matrilineal inheritance (hereafter, matriliny) in the village. C_i is the reported perceived

[29] We replace $M_h = 0.1$ when $M_h = 0$ to take the ratio.

cost of talking to children, while the vector $X_{i,c,h}$ captures how the cost of talking to children varies with parent, spouse, child, and household level characteristics as described above. These characteristics include the child's age and gender. To capture how the cost may be influenced by cognitive ability, we control for i's performance on a cognitive task similar to solving Raven's matrices. Given previous literature's findings about the importance of socio-economic conditions for parental investments (e.g., Cunha et al. (2013); Boneva and Rauh (2018)), we include an asset-based index of wealth of the household.[30] We also include the number of children in the household, which may influence how much time a parent may have to devote to the recorded child, and for whether the child is the father's biological father.[31] We analyze the sensitivity of our results to different specifications of the cost function.

We omit the province subscript for simplicity, but we control for a set of province fixed effects throughout our analysis. Speech was recorded by two USBs for each child, as insurance against the possible technical defects of a single recorder. To account for multiple recordings, we cluster standard errors at the child level in our baseline specifications, but consider alternative group-level structures of the error terms at the household or village level. Different USBs also had different performances due to random technical or accidental factors, so that the duration of recording varies across observations. We expect this heterogeneity to affect the precision of our estimates, so we weight each observation by the recording duration of each USB.

Key to our identification strategy is the heterogeneity in expected returns within couples. Without this variation, it would be impossible to disentangle the role of individual beliefs from those of their spouses, and consequently, to understand how bargaining power shapes the elasticity of investment to spousal beliefs.

We can identify the cost parameters $\kappa_{1,i}$ and γ_i, and the ratios of other structural parameters from estimating Equations (12) and (13), in particular $\delta_{1,i} = \frac{\omega_i}{\chi_{i,2}}$, $\delta_{3,f} = \beta * \frac{\omega_m}{\chi_{f,2}}$ and $\delta_{3,m} = \frac{1}{\beta}\frac{\omega_f}{\chi_{m,2}}$. While we cannot identify the elements of these ratios separately, this is not necessary for our counterfactual simulations that rely on shifting the beliefs $\eta_{i,2}$, the cost C_i, or matrilineal inheritance M_h.

The theoretical prediction is that $\delta_{1,m}$ and $\delta_{1,f}$ are positive since they capture parental

[30] The index is defined as the sum of dummy variables taking value one if the household owns a: mobile phone, radio, iron roof, solar light, water tank, toilet, stove, and outboard motor.

[31] Since our fieldwork took place in the context of the roll-out of an RCT about child-directed speech, which targeted either mothers or fathers, we control for treatment assignment in all specifications. Treatment assignment is balanced across matrilineal and patrilineal villages, and was implemented after we measured subjective beliefs and costs. Our results should thus be unaffected by the intervention. When regressing matriliny prevalence on our treatment conditions, we obtain a P-value for the coefficient associated with the mother treatment condition of 0.801, and 0.986 for the father treatment condition, relative to the control condition. We were unable to collect follow-up data on the intervention because of COVID and the country's shutdown.

preferences for child development (scaled by the positive cost parameter). The parameters $\delta_{3,f}$ and $\delta_{3,m}$ are thus expected to be positive as well. We are primarily interested in $\delta_{3,f}$ which measures how the elasticity of father's investment to mother's beliefs varies with women bargaining power. Finally, we also expect κ_1, which measures how the perceived cost of talking to children, influences the investment, to be negative.

5 Results

We first document basic differences in investments across mothers and fathers by estimating a simpler version of Equations (19) and (20) in which we only consider how total vocalizations per day vary based on *own* beliefs and costs parameters. The results are presented in column 1 of Table 3 (top panel for mothers and bottom panel for fathers). The results reveal an asymmetric pattern across genders. For mothers, only the coefficient associated with their expected beliefs about the returns of speech is statistically significant, while for fathers, only the coefficient associated with their self-reported direct cost of speech is statistically significant.[32] The coefficient associated with expected beliefs ("*Returns*") measures $\frac{\omega_i}{\chi_{i,2}}$ for parent i. Our results thus suggest that the ratio of preference for language development to cost parameter $\chi_{i,2}$ is a significant driver of investments for mothers, but not for fathers. In other words, mothers' investments are more elastic to beliefs, while fathers' investment are more elastic to direct costs. For mothers, a one standard deviation increase in returns is associated with a 0.11 standard deviation increase in vocalizations. For fathers, a one standard deviation decrease in direct costs is associated with a 0.11 standard deviation increase in vocalizations.

Our model also predicts that parental investment increases with their spouse's bargaining power. While we do not observe a statistically significant effect of male bargaining power on maternal investment (the coefficient on 1/Matriliny is positive but imprecisely estimated), we find a substantial effect of female bargaining power on fathers' total vocalizations per day. Specifically, transitioning from a fully patrilineal to a fully matrilineal system increases fathers' vocalizations by 0.60 standard deviations. This result aligns with the descriptive evidence presented in Table 2.

Column 2 of Table 3 augments column 1 by considering the role of spouse's expected returns. The coefficients associated with spouse's returns are small and statistically insignificant, both for mothers and fathers. Spousal beliefs about the returns to parental speech thus do not influence individual investment, on average.

[32]The difference between the coefficients associated with maternal and paternal beliefs is statistically significant at the 12% level. The difference in the costs coefficients is statistically significant at the 7% level.

Columns 3 presents the estimation results of Equation (19) and (20) that matches our structural model. We focus on the elasticities of parental investments to their spouse's expected return. For mothers, the coefficients associated with fathers' returns are insignificant, regardless of bargaining power. The model predicts that when fathers care about language development, they will leverage their bargaining power to induce more investments by mothers. Yet, we have found evidence that fathers' investments are not elastic to their own returns, which suggests that fathers have low preferences for language development.[33] It is therefore not surprising that maternal investments do not respond to fathers' beliefs: even when fathers have high bargaining power, they do not seek to elicit maternal investments in the particular domain studied in this paper.

For fathers' investments (column 3, bottom panel), the coefficient on the interaction between mothers' expected returns and matriliny—which captures the elasticity of fathers' investment to mothers' beliefs in contexts where women have high bargaining power—is positive and statistically significant at the 5% level. This finding is consistent with our theoretical prediction that when women have greater bargaining power, their beliefs about the returns to investment will have a stronger influence on fathers' investments. The effect size is large. A one standard deviation increase in mothers' returns is associated with a 0.46 standard deviation increase in fathers' speech in matrilineal villages.[34]

Since we consistently find a positive and significant coefficient associated with the elasticity of maternal investments to mothers' returns, this additional result suggests that mothers have a strong preference for language development and leverage their bargaining power to elicit higher investment by their husbands, when they believe parental investments are more productive.

For more flexibility, column 4 augments column 3's specification to allow for own preference and direct cost parameters to vary by bargaining power shifter. The results are similar. If anything, the main coefficient of interest associated with the interaction between matriliny and mothers' returns in driving paternal speech investments is larger in magnitude and more precisely estimated. As shown in column 1 of Table B5, the results are also similar in magnitude and statistical significance when we do not include any parent, spouse, child, and household level characteristics as cost shifters. Tables A13 and A14 display all the coefficients associated with all the included cost shifters.

Since the evidence thus far suggests that maternal investments are only driven by maternal returns, we focus hereon on paternal investment and how it varies by matrilineal

[33] To be precise, this evidence is consistent with a low $\frac{\omega_f}{\chi_{f,2}}$.

[34] As a comparison, Dupas et al. (forthcoming) finds that informing mothers in Ghana about the benefits of communication through a short video increases the number of male adult words by 0.51 standard deviation one day after the intervention.

inheritance. Nevertheless, given that our model also confirms that measurement errors are correlated across maternal and paternal investments,[35] we maintain our joint estimation strategy.

We now examine other dimensions of paternal investments: the quantity and quality of child-directed speech. Results are in Table 4. Even columns display the more flexible specification, as in column 4 of Table 3. Odd columns omit parent, spouse, child, and household characteristics as cost shifters. The coefficient associated with the interaction between mothers' returns and matrilineal inheritance in predicting conversational turns is positive but imprecisely estimated, which may be due to measurement errors in the CTC measure (see discussion in Section 3.4).

We improve on this measure by considering the number of vocalizations at high pitch as an additional proxy of the quantity of child-directed speech in columns 3 and 4, and the pitch during a conversational turn as a measure of the quality of child-directed speech in columns 5 and 6. The coefficient associated with the interaction between mothers' returns and matrilineal inheritance is positive and statistically significant at the 5% level in predicting both measures, suggesting high elasticity of paternal quantity and quality of child-directed speech with respect to mother's returns when she has more bargaining power. A one standard deviation increase in mothers' returns is associated with a 0.37 standard deviation increase in the number of vocalizations at high pitch and a 0.45 standard deviation increase in fathers' pitch during a conversational turn in matrilineal villages. These results suggest that the paternal speech induced by mothers with higher returns in matrilineal villages is directed to the child and is more engaging to the child.
Figure A2 shows evidence of good model fit for both parents and all speech outcomes.

6 Robustness

We provide a battery of robustness analyses. First, we discuss the quality of our probabilistic beliefs data and the robustness of our results to alternative modeling choices for the beliefs and to potential measurement error in Appendix B.1. In particular, Table B2 demonstrates that our conclusions remain unchanged after excluding the small number of respondents who violate the monotonicity of probabilities in practice questions, incorporating observations with negative returns to investment in speech, excluding observations with missing returns or costs, and addressing potential measurement errors in beliefs by averaging across our primary measure of returns to speech and alternative metrics, following methodologies outlined in Cunha et al. (2020) and Bhalotra et al. (2024).

[35] The ξ^2 of a Breusch-Pagan test of independence in column 4 is 90.83 (P-value: 0.00)

We also illustrate our results in the raw data by plotting the distribution of fathers' vocalizations according to matrilineal inheritance and mothers' expected returns. Figure A3 shows that the median, 25th, and 75th percentiles of fathers' vocalizations are higher for men married to high returns mothers in matrilineal villages, where we expect the investments by fathers to be higher. Median vocalizations by fathers married to high returns mothers in matrilineal villages are one standard deviation higher than median vocalizations in any other group (high returns mothers in *pa*trilineal villages, low returns mothers in matrilineal villages, low returns mothers in patrilineal villages). The distribution of the raw data thus suggests that our main results are not driven by outliers.

We also show that there is considerable within-village variation in beliefs and costs, alleviating potential concerns that our relationship of interest is driven by clustering of beliefs or costs in certain village types, rather than by parents acting on their beliefs and costs. Figure A4 depicts a box plot of mothers' and fathers' beliefs and costs for each village in our sample ordered by inheritance type (from patrilineal to matrilineal). The Figure shows considerable within-village variation in beliefs and costs across all types of inheritance practices. Table A6 shows that there is actually more within-village than between-village variation in the beliefs and costs. Overall, this alleviates the concern that our results may be driven by unobserved heterogeneity across villages in the distributions of beliefs and costs. Finally, we find very low correlation between beliefs and cost (0.0126 for mothers and 0.0391 for fathers), which goes against the idea that parents who anticipate higher returns for an investment internalize the cost of the investment and do not view it as costly.

We then consider robustness to alternative assumptions about Pareto weights and about the agents' utility function. In particular, we consider different functional forms for the relationship between Pareto weights and matriliny, and we relax the linearity assumption $\frac{\alpha}{1-\alpha} = \beta M_h$. First, we allow matriliny prevalence to flexibly shift the bargaining power by using categorical variables for each inheritance system. The results are displayed in Figure 2. The Figure plots the interaction coefficient between beliefs and matrilineal inheritance for different degrees along our scale of matrilineal inheritance, for vocalizations (panel (a)), vocalizations at high pitch (panel (b)), and pitch (panel (c)). For all measures, the increase in paternal investments driven by maternal beliefs increases monotonically with the degree of matriliny, suggesting our results are not driven by the specification choice of the bargaining power variable. Second, in Section B of the Appendix, we consider alternative specifications of the relationship between $\frac{\alpha}{1-\alpha}$ and M_h, including a non-linear logistic specification. We show that under some of these alternative specifications, the preference parameters are overidentified and we estimate them jointly out of the two optimal investment equations. Results in Tables B3 and B4 show that the coefficients capturing the elasticity of paternal

speech to mother's return in matriliny are statistically significant for all measures of paternal investments, including conversational turns.

We also make a series of different assumptions about the cost function. In Table B5, we first show that our results on total vocalizations, vocalizations at high pitch, and average pitch are insensitive to the choice of conditioning variables entering the cost function. Column 1 shows that the results are almost identical in specifications with no child or parent-level variables entering the cost functions. Column 2 further shows that all results are robust, and similar in magnitude to our baseline specification (shown in column 3), when we include child level variables only. In Column 4, we allow for all the factors to enter the cost function differently by matriliny. The results are unchanged. As shown in column 5, the results are robust when the cost also accounts for the amount of care that the child receives separately from female and male caretakers other than their parents.[36] To address the possibility that unobserved heterogeneity across villages or linguistic differences may explain our results, we show in column 6 that our results are robust to controlling for a set of 13 language fixed effects. Our coefficients of interest are robust and close in magnitude to our main estimates. In column 7, we show that our results are robust to accounting for the possibility that parents' costs may depend on the day of the recording by conditioning on date fixed effects. Column 8 shows that our results are unchanged when we include father's income source in the cost function, which may affect his opportunity cost. Last, in column 9, we consider vocalizations by children other than the key child in the cost function. Inputs by other children may crowd out or substitute for parental investments. Again, our results are unchanged.

Table B6 also shows that our results are robust to considering altruistic individuals who internalize their spouse's cost of parental speech, and we allow the altruism parameter to vary with matriliny, accounting for the possibility that matriliny may alter spousal altruism (Lowes, 2022).

Last, we consider different assumptions about the structure of the error terms. To deal with the few cases in which more than one child was recorded in the household, column 1 of Table A7 shows that our results are identical when we cluster standard errors at the household – rather than child – level, accounting for the possible correlation of the error term across siblings in the same household. In column 2, we revert to clustering standard errors at child level but keep only the child closest to two years old in the estimation sample, in line with our recruitment protocol. In the case of twins, we randomly drop one child. The

[36]The P-value in Panel B is exactly 0.100. Considering care by other caretakers also account for the possibility that the recorded vocalizations may not be from the child's parents. The results are robust and close in magnitude to our main estimates, suggesting that such potential measurement error does not bear on our results.

coefficient of interest becomes larger in magnitude. The last column of Table A7 shows that our results are robust when standard errors are clustered at the village level to account for the possibility of a group structure of the error term within matrilineal villages. The results are more precisely estimated for the quantity of vocalizations and child-directed speech.

7 Mechanisms

We now provide more evidence on the mechanisms through which mothers with more bargaining power are able to elicit paternal investment when they have high returns. We first show that paternal investment is specifically tailored to the domain in which high bargaining power mothers expect high returns and to their preferences. We then single out the role of women's *economic* bargaining power, as opposed to other potential dimensions of female status and show additional evidence on the role of women's *individual* economic bargaining power.

7.1 Mothers' beliefs as the driver of investment

Here, we show that higher (and better) paternal investments are specific to the domain where women perceive such investments as productive and align with women's child gender preferences. First, we pinpoint the role of maternal beliefs as opposed to other *maternal*, *paternal* and *child* characteristics that could influence paternal investments.

7.1.1 Beliefs vs. other determinants of investments

A potential concern is that maternal beliefs may proxy other variables that may systematically drive paternal speech. For example, women with higher returns may systematically marry spouses who would speak more to the child anyway (e.g., because they have low cost or high return); or they may have children who are able to elicit more paternal speech, for example, because they are better at engaging in conversations themselves.

Because we identify the parameter of interest through the interaction of beliefs and matriliny, this would only be a concern for our analysis if these potential confounders systematically varied with matriliny. Yet, as discussed in Section 3.3, we do not observe any systematic relationship between matrilineal inheritance and the spousal correlations in returns and costs. We also do not observe any systematic relationship between matrilineal inheritance and spousal correlations in formal education (P-value: 0.48) or cognition (performance on Raven-like matrices, P-value: 0.88). Moreover, high-returns matrilineal women marry men that are observationally similar to other fathers in terms of returns and costs of

child-directed speech, as well as age, formal education, income source, leisure, social preferences, and who they get help from with childcare. This is shown in Table A8, in which we regress fathers' characteristics on mothers' returns, matriliny, and the interaction between matriliny and mothers' returns.

Similarly, Table A9 shows that the *children* of high-returns matrilineal women are also observationally similar to other children in terms of BMI and gender (columns 1 and 2), suggesting that child characteristics may not drive our results. We also find that high-returns matrilineal women *themselves* are not observationally different from other women, along a wide range of characteristics, such as age, education, income source, leisure, number of children in the household, or social preferences (columns 3 to 11). Last, their children are not more likely to receive care from grandmothers or non-relatives (as reported both by mothers, as shown in columns 12 to 14 of Table A9, and by fathers, as shown in columns 10 to 12 of Table A8).

To gauge the possible influence of other, unobservable dimensions of the match, or of the couple's child, we perform a randomization inference test, in which we randomly reassign the mother's returns over 1,000 repetitions. In other words, we keep all other dimensions of the match between a husband and wife as well as the child's characteristics constant, and only systematically vary the mother's returns. Figure A5 displays the results for all three outcomes of interest, and shows that our estimated coefficients of interest on the interaction between mother's belief and matriliny are well outside the range of coefficients estimated from the random reassignments.

A last potential concern is that higher beliefs among women may reflect learning by those who place a strong value on child development. If women with a higher taste for child language development engage in more frequent communication with their children, they might learn that speech has high return. This could lead to a positive correlation between the unobserved taste and the expected return. However, our findings do not provide evidence of learning. Parents with a greater number of children do not exhibit systematically different expected returns (see Table B1), and the correlation of returns for parents, who observe the development of the same children, is low (refer to Figure 1). Moreover, to affect our results, learning should be heterogeneous by matriliny, which is unlikely given the absence of differences in mother investment with matriliny.[37]

[37]Finally, even if the positive coefficient of the interaction between the expected return and matriliny were entirely driven by an omitted taste parameter, our results would indicate similar intrahousehold dynamics to the ones we have described: powerful women who value parental investment can induce greater paternal investment.

7.1.2 General versus specific investments

Overall, these additional analyses suggest that paternal investment in matrilineal places is specifically driven by maternal beliefs. A question, however, is whether such paternal investment is specific to the domain of maternal beliefs, or only trickles down from a higher level of general paternal care, since more powerful women may simply be able to extract more care from the father. To answer this question, we collected data on father's time spent on child care, as reported by mothers.[38] Column 1 of Table 5 confirms that, in line with standard models of household bargaining, fathers spend more time taking care of their children in matrilineal villages. However, as shown in column 2 of Table 5, this measure of overall time investment does not vary with maternal beliefs in matrilineal villages. In other words, although all fathers spend more time taking care of their children in matrilineal villages, only those partnered with high returns mothers invest more in child-directed speech. Moreover, we show that all our results on speech investments are robust to conditioning on time spent by fathers taking care of children (Table A10). This further shows that our speech measures capture targeted investments rather than overall care and that paternal investments respond specifically to the domain of maternal beliefs.

Last, we show that paternal investments also align with typical maternal preferences over child gender. Existing literature typically shows that mothers have balanced preferences for boys and girls, while fathers tend to exhibit son's preference (e.g. Dizon-Ross and Jayachandran (2023), Lundberg (2005)).[39] To test which child (by gender) is the recipient of paternal investment, we augment Equation (20) with a triple interaction of maternal returns, matriliny and female child to allow for the preference parameter for language development to differ by child gender. The results, shown in columns 3 to 5 of Table 5 show that paternal speech induced by high-beliefs mothers in matrilineal villages is primarily directed toward girls. The coefficient associated with the triple interaction is positive and precisely estimated for total vocalizations, although less precise for other measures. Appendix Table A12 shows that the results are also significant for total vocalizations at high pitch when we consider a dummy variable taking value one in purely matrilineal villages.

[38] Women are asked: "How often does your husband help you taking care of your children?".

[39] In our sample, mothers display equal preferences for daughters and sons according to measures of maternal investment and preferences such as breastfeeding (Jayachandran and Kuziemko, 2011). In a regression where the dependent variable is the age at which mothers stopped breastfeeding on child, the coefficient associated with child female gender is 0.11 (p-value: 0.899). When interacted with matrilineal inheritance, it is 2.09 (P-value: 0.397).

7.2 Mothers' economic power as the driver of investment

In this section, we establish that our results are driven by female economic bargaining power, as opposed to other dimensions of female status.

7.2.1 Economic power vs. other dimensions of social status

Matrilineal inheritance may not just capture economic bargaining power, but also more general power or status of women, which itself could drive our results. To isolate the role of economic power from other dimensions of female status and power, we leverage the fact that our setting displays variation along both inheritance and residence patterns.

Under matrilocal residence patterns, newly married couples reside in the bride's village after marriage (see, e.g. Bau (2021)). Women have closer access to their kin in matrilocal places and have higher social status (Khalil and Mookerjee, 2019), but not necessarily more land or assets that would confer economic bargaining power. 15% of the villages in our sample are matrilocal (with the remaining 85% patrilocal), about half of which are also primarily matrilineal (M_h more than 0.5). This variation enables us to separate the influence of status and kinship networks from economic bargaining power. Column 1 of Table 6 reports the estimation results of Equation (20) replacing M_h by a dummy variable taking value one if the village is matrilocal. In contrast with matrilineal inheritance, we do not observe more father vocalizations in matrilocal villages, even when mothers have high returns: the coefficient associated with the interaction term of interest is negative and statistically insignificant. Moreover, in column 2, we allow female bargaining power to come from both matriliny and matrilocality by conditioning on both matriliny and matrilocality and their interactions with beliefs and costs. We find that only the interaction between matrilineal inheritance and maternal beliefs is significant. Columns 3 to 6 show similar results for the other quantitative and qualitative measures of fathers' speech investments.

Finally, in Table A11, we rely on drivers of female employment, such as the presence of a market, kindergarten, or primary school in the village,[40] which could influence female empowerment across villages. As before, we allow female bargaining power to come from both matriliny and from these additional village characteristics. In every case, the coefficient of interest associated with the interaction between matriliny and maternal beliefs remains statistically significant at the 1 to 5% level and is stable (often, larger) in magnitude. In the last column, we explore the role of the governance structure of the village, which could affect female political empowerment (presence of an elected leader vs. alternative governance

[40]The presence of a kindergarten or school could also influence child cognitive and non-cognitive skills as well as child social preferences, as shown by e.g, Heckman (2000); Cappelen et al. (2020), and thereby potentially influence parental investments.

structures such as hereditary chief, religious leader, big man, or council of elders). Our main results are again robust and stable in magnitude.

Overall, these results suggest that our results are specific to female economic bargaining power, rather than other sources of status or power stemming from the close presence of kin, as the driver of our results.

7.2.2 Direct evidence on economic power

We now offer direct evidence demonstrating the influence of economic power. First, as discussed in Section 6, we observe that the increase in paternal investments driven by maternal beliefs increases monotonically with the degree of matriliny – i.e., the intensity of female economic bargaining power.

Second, we provide evidence on the role of *individual* female economic power in columns 7 to 12 of Table 6. In columns 7 to 9, we first leverage variation stemming from quasi-random variation in parental death, which generates variation in whether women have already inherited land. We re-estimate Equation (19) replacing M_h by an individual level variable that takes value one if the woman has inherited land. Relying on such individual level variation enables us to augment our specification with village level fixed effects. In other words, we compare, within a village and traditional inheritance structure, women who have high *realized* economic bargaining power because land has already passed on to them vs. women who have not inherited. The results show that higher quantity of paternal investment is driven by high returns women who have already inherited.

In columns 10 to 12, we rely on another proxy of individual female economic bargaining power, which derives from marriage practices. The practice of bride price – where the groom's family transfers wealth to the bride's family at marriage – has been associated with higher educational investments in girls (Ashraf et al., 2020) and higher bargaining power and more autonomy conferred to women (Anderson, 2007; Lowes and Nunn, 2018), including in Melanesia and the Solomon Islands specifically (Guo, 2020; Brandl and Colleran, 2024).[41] In our sample, 34.23% of women have received a bride price at marriage.[42] The practice of bride price is uncorrelated with the prevalence of matrilineal inheritance at the village level (raw correlation: 0.005) but more prevalent, although not universal, in patrilocal villages (raw correlation: 0.28), consistent with the ethnographic record (Anderson, 2007; Brandl

[41] Brandl and Colleran (2024) also discusses potential drawbacks of bride price for women, mostly due to male resentment over high payments and higher cost of divorce for men who might find it difficult to save enough for a subsequent bride. Overall, however, they conclude that the ethnographic record is consistent with the benefits to women, in terms of status and economic security, outweighing the potential drawbacks. They also highlight that women themselves desire bride price.

[42] For the remaining 63.76% marriages, no exchange of money or goods took place, and in 2.01% of the cases, there was a reciprocal gift exchange.

and Colleran, 2024). Importantly, we rely on individual-level variation in bride price, thus comparing women who received a bride price to others in the same village, enabling us to abstract from other potential village-level determinants of female status. Columns 10 to 12 of Table 6 show, consistent with previous literature documenting higher female economic security and bargaining power associated with bride price, that the husbands of high returns women who have received a bride price talk more to their child and do so at a higher average pitch, with the measures of total vocalizations and pitch statistically significant at the 5% level.

8 Policy Experiments

We use the estimated structural parameters to simulate the behavioral responses of mothers and fathers to a series of policy interventions targeted at increasing the quantity of speech investment. These include manipulation of expected returns, perceived cost of talking to children, and bargaining power.[43] The simulations assume full compliance (e.g., all women fully revise their expectations). The results therefore constitute upper bounds of policy treatment effects. The estimates are presented in Table 7. Column 1 shows the baseline distributions of investments predicted by model (column 3 of Table 3 and Table 4) in fully matrilineal and fully patrilineal villages before any of the policies is introduced.

We first discuss the impact of **information interventions** on the benefits of verbal engagement, similar to what is implemented by Dupas et al. (forthcoming) and Cunha et al. (2024). These interventions can be thought of as delivering information about the objective production function (Equation 3), in particular about μ_2 which captures the perceived elasticity of child language development to parental investments. Consequently, the subjective production function (Equation 4) is revised, specifically with regards to $\eta_{i,2}$.

We benchmark the information intervention against the change in beliefs achieved in previous interventions. Based on the treatment effect on parental beliefs from Cunha et al. (2024) and from intensive parenting interventions (between 9 and 17 sessions) in low-income countries reviewed in Bhalotra et al. (2024), we consider interventions that lead to an increase in expected returns of mothers or fathers of 0.40 of a SD.

In column 2 of Table 7, we estimate that shifting maternal beliefs raises the predicted male vocalizations by 6% in fully matrilineal villages, has no effect on male vocalizations in fully patrilineal villages, and has modest positive effects on female vocalizations in either village. Effect sizes are similar for the number of high-pitched vocalizations. Overall, shifting maternal beliefs has modest effects on mothers, who already speak a lot, and only affects

[43] The policy interventions have minimal impact on pitch, so we focus on quantity.

fathers if mothers have high enough bargaining power.

Column 3 shows that, consistent with our previous results, shifting paternal beliefs has no effect on any of the parents' verbal investment with children.

We next simulate results of **reducing fathers' costs of talking** by 0.4 of a SD (column 4).[44] Such an intervention could for example consist in having fathers join a play group and learn fun activities to share with their children. Reducing fathers' costs is associated with a very modest (1-2%) increase in fathers' total and high-pitched vocalizations, with a comparable effect size across matrilineal and patrilineal villages.

The next simulation investigates impacts of a female empowerment intervention. We consider an intervention that is equivalent to shifting fully patrilineal inheritance to equal inheritance ($M_h = 0$ to $M_h = 0.5$). As shown in column 5, female empowerment would lead to a 17% increase in fathers' vocalizations (total and high-pitched), while mothers slightly reduce their investments in patrilineal places. Overall, the effect of this female empowerment policy on fathers' investments is much larger than either increasing mother's returns or decreasing father's costs.

We then combine the mother's beliefs, father's costs, and female empowerment policies in column 6. This combined intervention increases fathers' vocalizations by 21 to 22% in patrilineal places and 6 to 8% in matrilineal places.

9 Conclusion

This paper studies the intra-household allocation of verbal investments in children. We documents large gaps in speech investments in children between mothers and fathers. Women talk more overall, and use higher pitch when talking to their children. Yet, we show how female bargaining power can tilt this division of parental investment. Men invest more in child speech, and do so in a more engaging way for the child, when their wives believe investments are important determinants of child language development and when they have higher bargaining power. We show that these results are unlikely to be driven by assortative matching between parents, learning, or persuasion, by other characteristics of parents and children, or by unobservable village, partners, or child level differences. Paternal investment does not simply reflect higher paternal care but specific investment that matches the mother's beliefs. We provide further evidence, included at the individual level, that singles out the role of female economic bargaining power from other potential dimensions of female status. Altogether, our results suggest that more powerful women are able to elicit from their spouses

[44] We focus on fathers' costs since our results show that mothers' costs do not influence parental investments. As for fathers' returns, shifting mothers' costs has no effect on parental investments.

the investment that they believe matters.

Our findings have several implications for policy. We show that women, when deciding on their own parental investments, act on their beliefs about returns; while men act on their costs. Men also act on their wives' beliefs – but only when women are powerful enough. Consequently, our policy simulations show that lowering costs for men, for example through training about playful activities they might enjoy with their children, is effective in raising paternal investments, but modifying fathers' beliefs, for example with information treatments on the benefits of child-directed speech, is not. For mothers, reducing costs is not effective at changing their investments, while increasing their beliefs about returns has not only a direct effect on their own investments but may also have an indirect effect, through the elicitation of investments by their husbands, especially when coupled with women's empowerment programs. Indeed, the greatest gains in paternal investments would be achieved through female empowerment programs. Overall, our results stress the importance of considering the dynamic of intra-household allocation of investments and spillovers across parents when designing and evaluating early childhood interventions.

Finally, our study demonstrates the effectiveness of a scalable and cost-efficient method for measuring paternal and maternal language inputs. This method, coupled with our approach of measuring beliefs and costs, provides a comprehensive framework for understanding parental investment decisions. We hope this work inspires future research to adopt similar data collection strategies, furthering our understanding of the economics of parenting and promoting rich, engaging, and fruitful parental investments in children.

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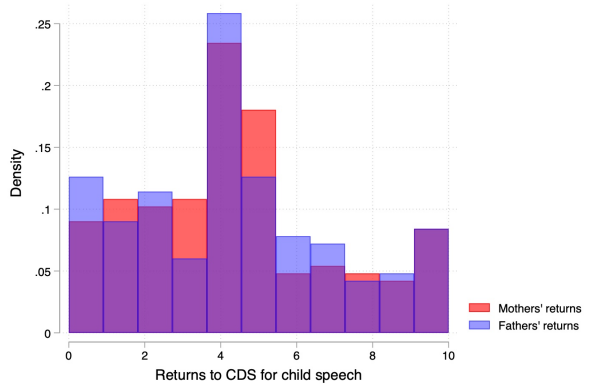
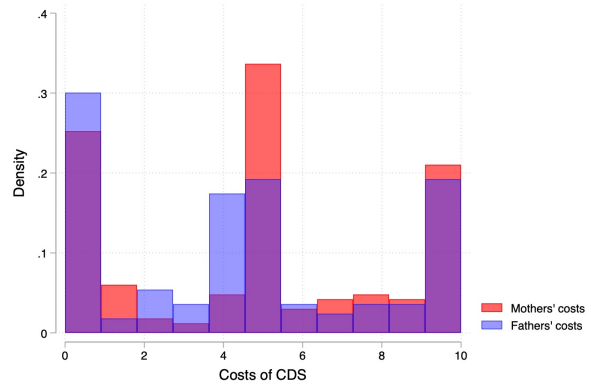
10 Figures

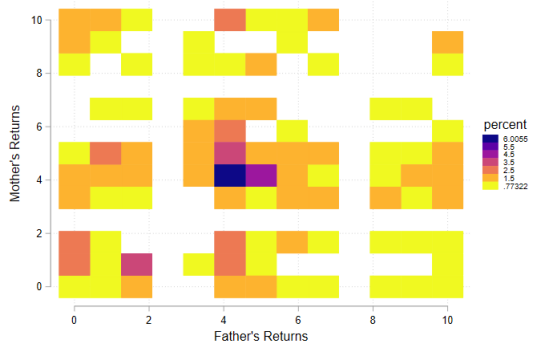
Figure 1: Mothers' and fathers' beliefs about returns and costs of Child-Directed Speech (CDS)



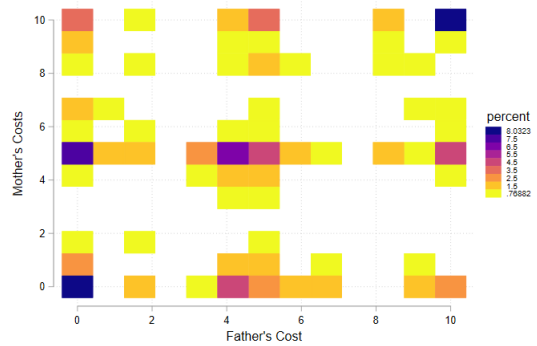
(a) Returns to CDS



(b) Costs of CDS



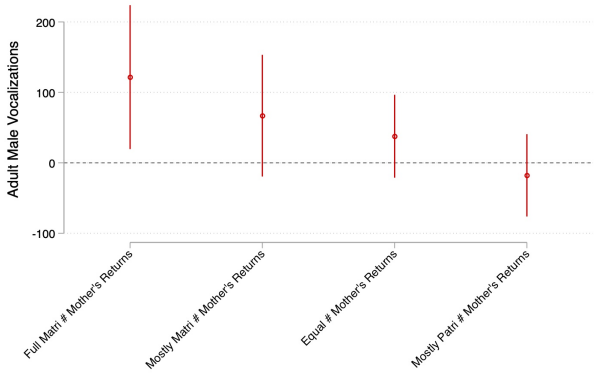
(c) Correlation of returns



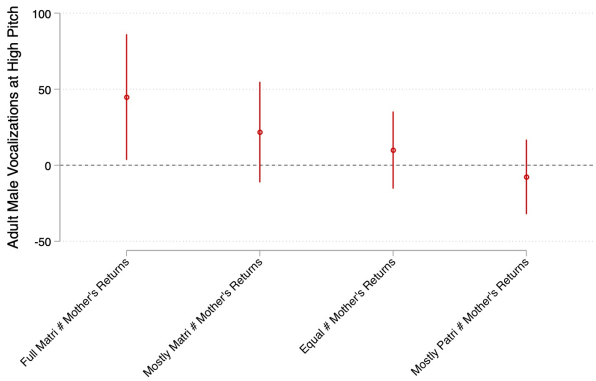
(d) Correlation of costs

Panel (a) display mothers' and fathers' expected return to talking to their child. A Kolmogorov-Smirnov tests of equality of the distributions has a p-value of 0.483. Panel (b) displays mothers' and fathers' costs of talking to their child. A Kolmogorov-Smirnov test of equality of the distributions has a p-value of 0.005. Panels (c) and (d) show a heat map of the correlation between mothers' and fathers' returns and costs, respectively.

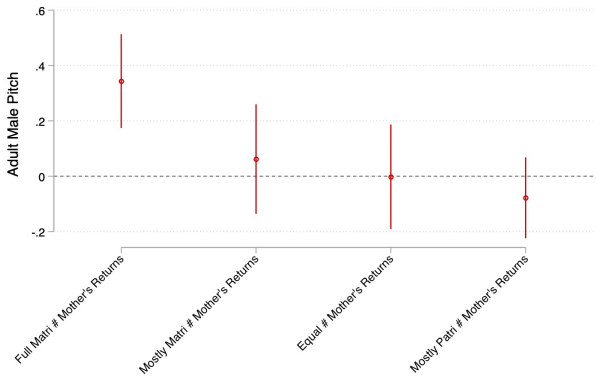
Figure 2: Main results by land inheritance category



(a) Vocalizations



(b) Vocalizations at High Pitch



(c) Pitch

The figures plot the joint estimation results of equations (19) when M_h is a categorical variable and displays father results. The dots correspond to the coefficients associated with the interaction between the mother's returns and each land inheritance type (as indicated). The vertical lines represent 90% confidence intervals. Full patriliny is the committed category. The dependent variable in Panel (a) is the total number of adult male vocalizations per day. The dependent variable in Panel (b) is the number of vocalizations at high pitch per day, where high pitch is above the speaker's average median pitch during conversational turns with the child. The dependent variable in Panel (c) is the average median pitch of the adult male speaker's voice during conversational turns with the child.

11 Tables

Table 1: Matriliny and female bargaining power

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Women	Inherit Land	Leisure	Age at Marriage	Share Sent to Spouse	Village Leadership	Generalized Trust	Compete with Spouse
Matriliny	0.119**	0.193**	1.733*	-0.105**	0.033	-0.213	-0.033
	(0.055)	(0.076)	(0.917)	(0.053)	(0.081)	(0.176)	(0.118)
Mean DV	0.90	0.83	22.08	0.47	0.48	2.49	0.58
SD DV	0.31	0.38	4.37	0.29	0.50	1.08	0.50
Panel B: Men	Inherit Land	Leisure	Age at Marriage	Share Sent to Spouse	Village Leadership	Generalized Trust	Compete with Spouse
Matriliny	0.037	0.127	0.308	-0.076	0.096	0.250	-0.118
	(0.063)	(0.079)	(1.204)	(0.055)	(0.089)	(0.260)	(0.116)
Mean DV	0.90	0.89	25.81	0.56	0.75	2.61	0.65
SD DV	0.30	0.31	4.80	0.27	0.43	1.26	0.48
Panel C: Household	Mobile Phone	Radio	Solar Light	Outboard Motor	Water Tank	Toilet	Stove
Matriliny	0.045	0.009	-0.009	-0.045	0.191**	0.236***	0.120
	(0.073)	(0.068)	(0.041)	(0.072)	(0.080)	(0.075)	(0.077)
Mean DV	0.73	0.21	0.93	0.27	0.57	0.34	0.32
SD DV	0.45	0.41	0.25	0.44	0.50	0.47	0.47

* p< 0.10, ** p<0.05, *** p < 0.01. OLS estimates. Each specification regresses the dependent variable indicated at the top of the column on the matriliny variable, controlling for a province fixed effect. The sample is restricted to people who have children, and is further restricted to women in panel A and men in panel B. In panel C, the unit of observation is the household. *Inherit land* is a binary variable equal to one if the respondent inherited land. *Leisure* is a binary variable equal to one if the respondent enjoys any leisure time. *Share sent to spouse* is the share of the dictator game endowment sent non-anonymously to the spouse. *Village Leadership* is a binary variable equal to one if the respondent is a member of any leadership organizations in their village. *Generalized Trust* is measured with the following question: *Generally speaking, would you say that most people can be trusted, or that you can't be too careful in dealing with people*, where 1 is complete distrust and 5 is complete trust. *Compete with Spouse* is a binary variable equal to one if the respondent chose to compete with their spouse in the compeitition task. The dependent variables in panel C are binary variables equal to one if the household owns the asset. Robust standard errors in parentheses.

Table 2: Matriliny and adult speech

Speaker:	Adult Female				Adult Male			
DV:	Vocs	Conv Turns	Vocs at High Pitch	Pitch	Vocs	Conv Turns	Vocs at High Pitch	Pitch
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Matriliny	-7.24	-127.78	-5.56	-0.20	356.77**	94.60*	172.04***	0.06
	(188.96)	(134.67)	(81.10)	(0.17)	(137.69)	(50.87)	(59.41)	(0.30)
Observations	377	377	377	373	377	377	377	371
R^2	0.098	0.179	0.127	0.151	0.118	0.243	0.177	0.170
# Clusters	196	196	196	196	196	196	196	196
Mean DV	2,782.89	1,415.36	1,163.20	28.66	1,312.15	441.93	583.84	22.76
SD DV	988.35	698.71	437.05	0.80	584.47	271.56	272.60	1.47

* p< 0.10, ** p<0.05, *** p < 0.01. OLS estimates. Unit of observation is a child recording. The dependent variable in columns (1) and (5) is the number of vocalizations per day. The dependent variable in columns (2) and (6) is the number of conversational turns with the child per day. The dependent variable in columns (3) and (7) is the number of vocalizations at high pitch per day, where high pitch is above the speaker's average median pitch during conversational turns with the child. The dependent variable in columns (4) and (8) is the average median pitch of vocalizations during a conversational turn. Pitch is missing if no conversational turn occurred during the recording. All estimates are weighted by the duration of the recording and control for the following variables: the log of child's age, number of children in the household, whether the mother's partner is the biological father of the child, the asset index, and a province fixed effect. Columns (1)-(4) control for mother's cognitive ability and columns (5)-(8) control for father's cognitive ability. Missing observations imputed for cognitive ability and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses.

Table 3: Returns, costs, and vocalizations

DV: Adult vocalizations per day	(1)	(2)	(3)	(4)
Panel A: Adult Female Speech				
Mother's Returns	36.171*	38.542*	38.370*	55.555
	(21.929)	(22.806)	(22.781)	(41.863)
Mother's Costs	16.798	16.240	13.526	23.327
	(16.374)	(16.437)	(16.329)	(28.189)
Father's Returns		-12.512	6.093	0.038
		(21.858)	(36.936)	(45.594)
1/Matriliny × Father's Returns			-2.921	-1.985
			(4.783)	(6.102)
1/Matriliny × Mother's Returns				-2.630
				(5.580)
1/Matriliny × Mother's Costs				-1.420
				(3.803)
1/Matriliny	28.014	28.426	41.864	54.533
	(17.291)	(17.477)	(30.132)	(52.206)
R^2	0.23	0.23	0.23	0.22
Mean DV	2,782.89	2,782.89	2,782.89	2,782.89
SD DV	988.35	988.35	988.35	988.35
Panel B: Adult Male Speech				
Father's Returns	-4.256	-7.865	-7.617	-3.203
	(12.842)	(14.521)	(14.868)	(17.574)
Father's Cost	-18.369*	-18.047*	-16.244*	-13.825
	(9.932)	(9.874)	(9.755)	(12.283)
Mother's Returns		3.699	-18.437	-20.191
		(16.421)	(18.793)	(18.816)
Matriliny × Mother's Returns			95.212**	101.509**
			(46.040)	(47.820)
Matriliny × Father's Returns				-15.558
				(52.053)
Matriliny × Father's Cost				-9.177
				(25.635)
Matriliny	354.039***	354.756***	-6.307	81.629
	(125.169)	(126.588)	(213.645)	(381.874)
Observations	377	377	377	377
# Clusters	196	196	196	196
R^2	0.15	0.15	0.18	0.18
Mean DV	1,312.15	1,312.15	1,312.15	1,312.15
SD DV	584.47	584.47	584.47	584.47

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother (Panel A) and father (Panel B) equations. Unit of observation is a child recording. All estimates control for the log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, the asset index, and a province fixed effect. Panel A additionally controls for mother's cognitive ability. Panel B additionally controls for father's cognitive ability. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

Table 4: Child-directed speech by fathers

DV:	Conv Turns		Vocs at High Pitch		Pitch	
Speaker: Adult male	(1)	(2)	(3)	(4)	(5)	(6)
Matriliny × Mother's Returns	8.270	5.544	38.743**	35.369*	0.236**	0.233**
	(15.228)	(14.540)	(19.408)	(19.356)	(0.102)	(0.103)
Matriliny × Father's Returns	-20.178	-21.304	-16.425	-16.026	-0.009	-0.013
	(19.430)	(19.315)	(22.911)	(22.634)	(0.085)	(0.086)
Matriliny × Father's Cost	1.118	-8.985	7.125	0.792	0.037	0.038
	(10.236)	(10.844)	(11.436)	(11.958)	(0.068)	(0.070)
Mother's Returns	0.846	-2.557	-3.656	-6.112	-0.072	-0.072
	(7.584)	(8.159)	(7.714)	(7.916)	(0.045)	(0.045)
Father's Returns	3.232	2.385	-2.106	-3.247	0.053	0.054
	(8.475)	(7.958)	(8.160)	(8.102)	(0.041)	(0.041)
Father's Cost	-6.322	-4.211	-7.399	-6.177	0.031	0.030
	(6.555)	(6.277)	(5.741)	(5.626)	(0.028)	(0.028)
Matriliny	137.768	222.383	54.025	105.656	-0.963	-0.949
	(143.000)	(145.088)	(165.310)	(170.096)	(0.644)	(0.637)
Controls:	N	Y	N	Y	N	Y
Observations	377	377	377	377	371	371
R^2	0.22	0.27	0.20	0.22	0.21	0.21
# Clusters	196	196	196	196	196	196
Mean DV	441.93	441.93	583.84	583.84	22.76	22.76
SD DV	271.56	271.56	272.60	272.60	1.47	1.47

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. Unit of observation is a child recording. The dependent variable in columns (1)-(2) is the number of conversational turns per day between the child and adult male speaker. The dependent variable in columns (3)-(4) is the number of vocalizations at high pitch per day, where high pitch is above the speaker's average median pitch during conversational turns with the child. The dependent variable in columns (5)-(6) is the average median pitch of the adult male speaker's voice during conversational turns with the child. Pitch is missing if no conversational turn occurred during the recording. All estimates control for a province fixed effect. Columns (2), (4), and (6) include the following controls: the log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, father's cognitive ability, and the asset index. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

Table 5: Child care and heterogeneity by child gender

DV:	Father Child Care		Adult Male Speech		
			Vocs	Vocs at High Pitch	Pitch
	(1)	(2)	(3)	(4)	(5)
Matriliny	0.355*	0.700	785.374**	333.596**	-0.827
	(0.212)	(0.546)	(357.955)	(163.954)	(0.796)
Matriliny × Mother's Returns		-0.007	1.992	5.257	0.198
		(0.075)	(46.934)	(20.994)	(0.128)
Female child × Matriliny			-923.362***	-256.101*	-0.958
			(353.015)	(151.655)	(0.769)
Female child × Matriliny × Mother's Returns			172.519**	45.373	0.159
			(76.216)	(30.985)	(0.166)
Observations	196	196	377	377	371
R^2	0.26	0.28	0.26	0.29	0.25
Mean DV	0.00	0.00	1,312.15	583.84	22.76
SD DV	1.00	1.00	584.47	272.60	1.47

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. In columns (1)-(2), the unit of observation is a household, and the dependent variable is how often the father takes care of the children, standardized to mean zero, standard deviation one. In columns (3)-(5), the unit of observation is a child recording. In column (3), the dependent variable is adult male vocalizations per day. In column (4), the dependent variable is the number of adult male vocalizations at high pitch per day, where high pitch is above the speaker's average median pitch during conversational turns with the child. In column (5), the dependent variable the average median pitch of the adult male speaker's voice during conversational turns with the child. Pitch is missing if no conversational turn occurred during the recording. All estimates control for log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, the asset index, father's cognitive ability, and a province fixed effect. Columns (3)-(5) also control for father's returns and costs, interacted with *matriliny*. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Robust standard errors in parentheses in columns (1)-(2). Columns (3)-(5) cluster standard errors at the child level and weight estimates by the duration of the recording.

Table 6: Female status and father speech investments

Measure of Female Status:	Matrilocality						Inherited Land			Bride Price		
DV: Adult Male Speech	Vocs	Vocs	Vocs at High Pitch	Vocs at High Pitch	Pitch	Pitch	Vocs	Vocs at High Pitch	Pitch	Vocs	Vocs at High Pitch	Pitch
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Female status × Mother's Returns	-51.768	-51.463	-14.722	-11.175	-0.145	-0.182	58.628**	22.151*	-0.063	79.920**	23.656*	0.152**
	(41.115)	(42.401)	(19.791)	(20.450)	(0.153)	(0.144)	(29.538)	(11.907)	(0.076)	(31.753)	(13.881)	(0.063)
Female status	-15.703	47.968	-26.291	-21.569	-0.997	-0.713	-376.955**	-172.108**	1.146**	-529.977**	-203.923**	-0.557
	(413.675)	(440.276)	(184.053)	(189.378)	(0.737)	(0.787)	(189.545)	(85.205)	(0.468)	(217.363)	(96.944)	(0.497)
Matriliny × Mother's Returns		101.620**		33.006*		0.251***						
		(49.619)		(19.574)		(0.096)						
Matriliny		62.010		100.945		-0.984						
		(397.575)		(169.139)		(0.700)						
Observations	377	377	377	377	371	371	377	377	371	311	311	305
R^2	0.12	0.19	0.18	0.23	0.24	0.26	0.40	0.44	0.43	0.40	0.45	0.49
Mean DV	1312	1312	584	584	23	23	1312	584	23	1316	592	23
SD DV	584.47	584.47	272.60	272.60	1.47	1.47	584.47	272.60	1.47	566.93	268.44	1.50

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. The unit of observation is a child recording. All specifications regress a measure of adult speech on a measure of female status interacted with mother's returns, father's costs, and father's returns, controlling for log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, the asset index, father's cognitive ability, and a province fixed effect. In columns (1)-(6), the measure of female status is whether the village is a matrilocal village. In columns (7)-(9), the measure of female status is whether the mother has inherited land. In columns (10)-(12) the measure of female status is whether the mother's family received a bride price at marriage and is restricted to children of married couples. *Vocs* is the number of adult male vocalizations per day. *Vocs at High Pitch* is the number of adult male vocalizations at high pitch per day, where high pitch is above the speaker's average median pitch during conversational turns with the child. *Pitch* is the average median pitch of the adult male speaker's voice during conversational turns with the child. Pitch is missing if no conversational turn occurred during the recording. All estimates cluster standard errors at the child level (in parentheses) and are weighted by the duration of the recording.

Table 7: Policy experiments

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Baseline	Increase mother returns by 0.4 s.d	Increase father returns by 0.4 s.d.	Reduce father costs by 0.4 sd	Empower women $M_h = 0$ to 0.5	Empower women, m's returns, f's costs (2) + (4) + (5)	Percentage change (6) - (1)
Male vocalizations							
Patri	1297.53	1274.78	1293.84	1317.14	1519.88	1580.44	21.80
Matri	1546.07	1637.70	1524.41	1578.69	1546.07	1670.32	8.04
Female vocalizations							
Patri	2944.30	2977.27	2921.43	2944.30	2760.38	2790.07	-5.24
Matri	2760.38	2820.01	2758.13	2760.38	2733.40	2820.01	2.16
Male vocs at high pitch							
Patri	585.21	578.33	581.47	593.97	685.86	707.10	20.83
Matri	681.22	714.19	658.97	688.85	681.22	721.82	5.96
Female vocs at high pitch							
Patri	1241.92	1263.02	1238.64	1241.92	1143.61	1166.44	-6.08
Matri	1137.93	1160.97	1138.11	1137.93	1137.93	1160.97	2.02

Predicted vocalizations estimated after joint estimation of mother and father equations. Vocalizations is the number of adult vocalizations per day. Vocalizations at High Pitch is the number of adult vocalizations at high pitch per day, where high pitch is above the speaker's average median pitch during conversational turns with the child. The vocalizations (top panel) are predicted from the coefficients presented in Table 3, column 3. The vocalizations at high pitch (bottom panel) are predicted from the coefficients presented in Table 4, column 4.

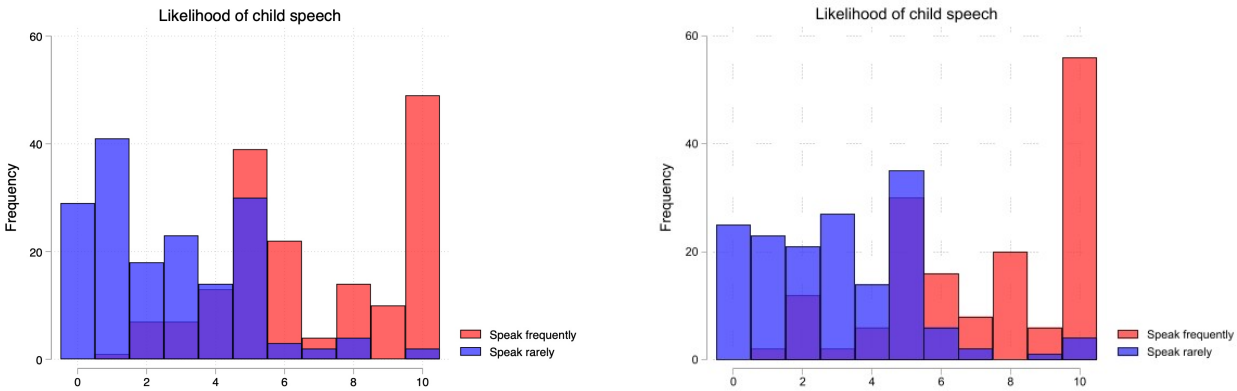
HeforShe: Bargaining Power, Parental Beliefs, and Parental Investments

Appendix

A Additional Results

A.1 Additional Figures

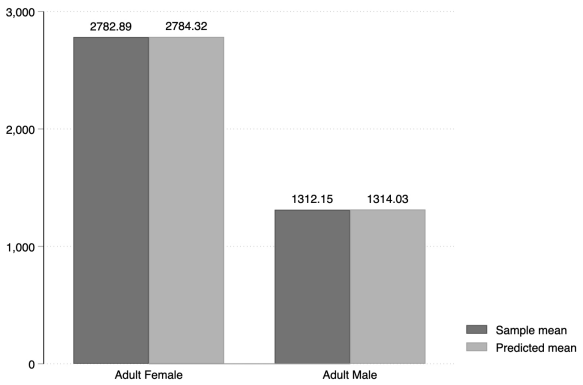
Figure A1: Distribution of beliefs



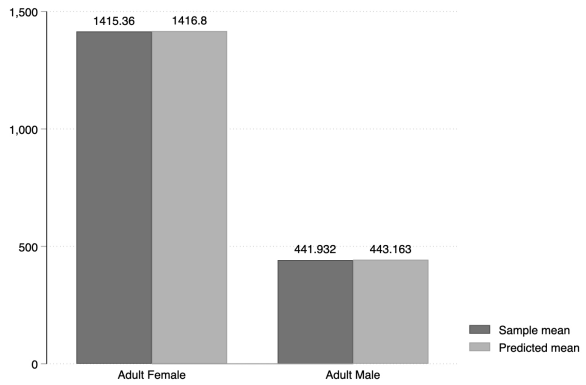
(a) Mother's Beliefs (b) Father's Beliefs

The Figure displays mothers' and fathers' beliefs about the likelihood that a child puts together 2-3 words by age 2 when the carer speaks frequently and rarely to the child.

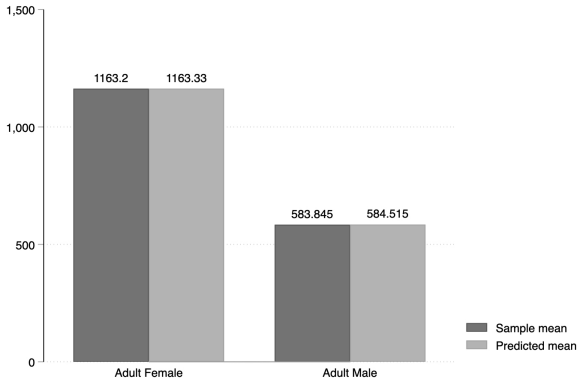
Figure A2: Model Fit



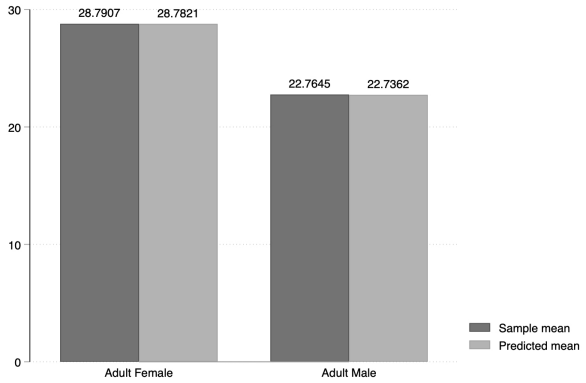
(a) Vocalizations



(b) Conversational Turns (CTC)



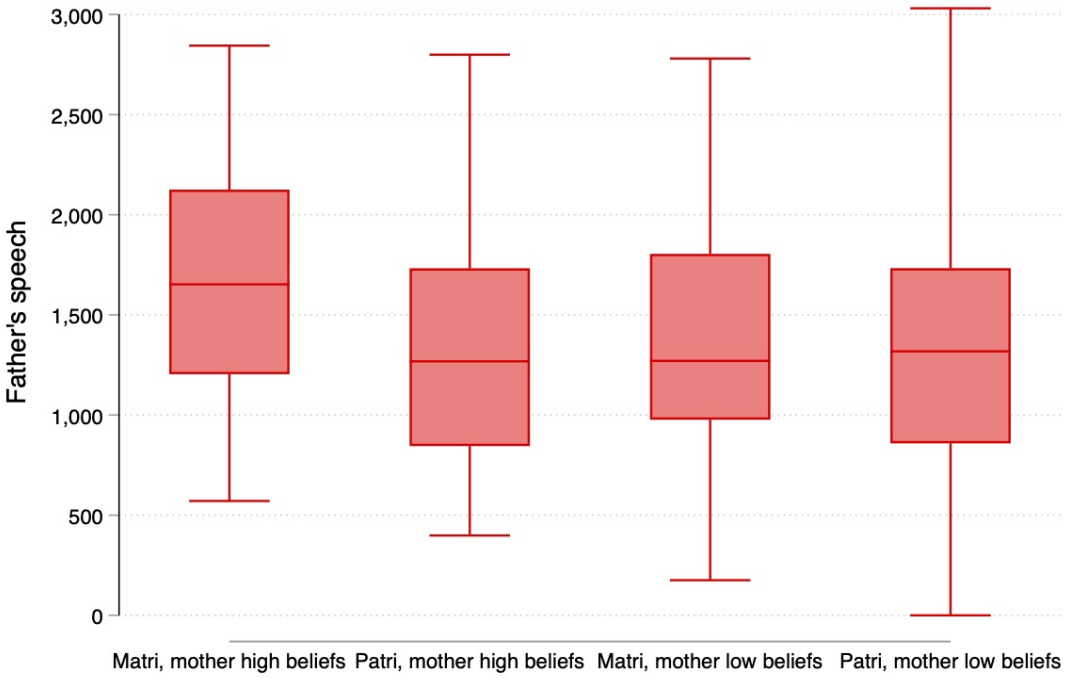
(a) Vocs at High Pitch



(b) Pitch

The Figure displays the sample mean (dark gray) and predicted mean (light gray) for each measure of speech. In Panel (a), the specification for *Vocalizations* comes from column (4) of Table 3. In Panel (b), the specification for *Conversational Turns (CTC)* comes from column (2) of Table 4. In Panel (c), the specification for *Vocs at High Pitch* comes from column (4) of Table 4. In Panel (d), the specification for *Pitch* comes from column (6) of Table 4.

Figure A3: Father's speech by mother's beliefs and matriliny



The Figure displays a box plot of fathers' vocalizations by matriliny and mothers' beliefs. We divide the sample into four groups according to matrilineal status of the village ("Matri": M_h equal to or above 0.75, otherwise "Patri") and mothers' expected returns ("high" is above median, otherwise "low"). The center line in each box represents the 50th percentile (median). The bottom of each box represents the 25th percentile and the top represents the 75th percentile. The interquartile range is the difference between the 75th and 25th quartiles. The bottom whisker below the box is equal to the 25th percentile minus 1.5 times the interquartile range. The upper whisker above the box is equal to the 75th percentile plus 1.5 times the interquartile range.

Figure A4: Village variation in beliefs and costs

(a) Mother's Returns

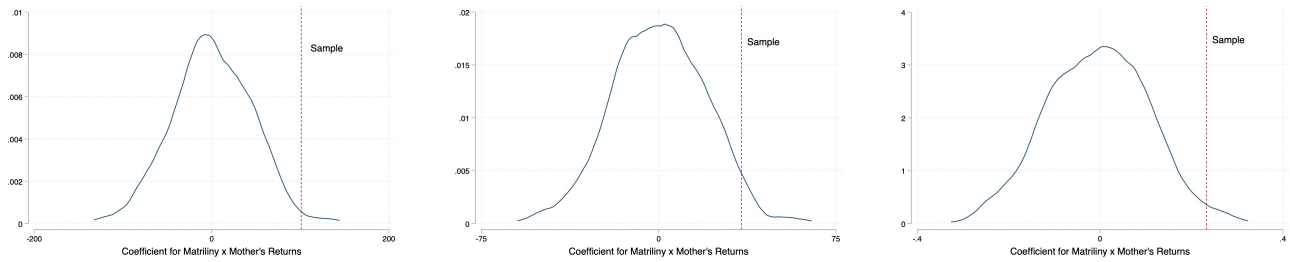
(b) Father's Returns

(c) Mother's Costs

(d) Father's Costs

The Figure displays the village-level box plots for mother and father returns and costs. Villages are sorted on the x-axis in order of matriliny, where villages 1-17 are purely patrilineal, 18-28 are mostly patrilineal, 29-31 are equally patrilineal and matrilineal, 32-37 are mostly matrilineal, and 38-41 are purely matrilineal.

Figure A5: Randomization inference - Adult Male Speech



(a) Vocalizations (b) Vocalizations at high pitch (c) Pitch during CTC

The Figure plots the results of randomization inference exercises that replicate Panel B, column (4) of Table 3 (in panel (a)), column (4) of Table 4 (in panel (b)), and column (6) of Table 4 (in panel (c)) by randomly re-assigning the mother's beliefs about the returns to CDS over 1,000 repetitions. The solid line plots the kernel density of the coefficient estimates for *Matriliny x Mother's Returns*. The vertical dash line represents the coefficients estimated from the true sample (see Panel B, column (4) of Table 3, column (4) of Table 4, and column (6) of Table 4).

A.2 Additional Tables

Table A1: Summary statistics

	N	Mean	SD	Min	Max
Household Variables					
Purely patrilineal	182	0.45	0.50	0	1
Mostly patrilineal	182	0.26	0.44	0	1
Equally patri and matri	182	0.07	0.25	0	1
Mostly matrilineal	182	0.12	0.33	0	1
Purely matrilineal	182	0.11	0.31	0	1
Mother's age	182	29.38	5.67	20	47
Father's age	182	33.40	6.00	21	54
Total children in HH	182	3.03	1.66	1	11
Father not biofather	182	0.16	0.37	0	1
Mother's cognitive ability	182	11.32	6.27	0	23
Father's cognitive ability	182	13.59	5.74	0	23
Mother completed secondary school	182	0.31	0.46	0	1
Father completed secondary school	182	0.35	0.45	0	1
Asset index	182	3.87	1.89	0	8
Mother's Returns	182	4.33	2.80	0	10
Father's Returns	182	4.32	2.91	0	10
Mother's Costs	182	4.88	3.58	0	10
Father's Cost	182	4.41	3.57	0	10
Earns income from employment	182	0.04	0.21	0	1
Access to grid electricity	182	0.07	0.25	0	1
Access to running water	182	0.03	0.18	0	1
Water tank	182	0.57	0.48	0	1
Child Variables					
Recorded child's age (months)	196	25.30	9.03	6	48
Recorded child female	196	0.47	0.50	0	1

Asset Index is defined as the sum of dummy variables taking value one if the household owns a: mobile phone, radio, iron roof, solar light, water tank, toilet, stove, and outboard motor.

Table A2: Compliance with recording

DV:	Comply (1)	Comply (2)
Matriliny	0.008	0.017
	(0.029)	(0.032)
Controls:	N	Y
Observations	377	377
R^2	0.003	0.032
Mean DV	0.95	0.95
SD DV	0.21	0.21

* p< 0.10, ** p<0.05, *** p < 0.01. OLS estimates. Unit of observation is a child recording. Dependent variable is equal to one if the duration of the recording is greater than the fifth percentile (302 minutes). All estimates control for a province fixed effect. Column (2) includes the log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, mother's cognitive ability, and the asset index. Missing observations imputed for cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses.

Table A3: Assortative matching in beliefs by matriliny

DV: Parental beliefs	Returns		Costs	
	|Mother-Father| (1)	|Mother-Father| (2)	|Mother-Father| (3)	|Mother-Father| (4)
Matriliny	-0.543	-0.499	-0.678	-0.491
	(0.565)	(0.566)	(0.698)	(0.704)
Additional Controls:	N	Y	N	Y
Observations	182	182	182	182
R^2	0.116	0.171	0.057	0.100
Mean DV	3.17	3.17	3.37	3.37
SD DV	2.46	2.46	2.95	2.95

* p< 0.10, ** p<0.05, *** p < 0.01. OLS estimates. Unit of observation is the parents of a recorded child. In each specification, the dependent variable is the absolute value difference in mothers' and fathers' returns (columns 1-2) and costs (columns 3-4). All estimates control for a province fixed effect. Additional controls in columns 2 and 4 include: child age, child gender, number of children in the household, whether the father is the biological father of the recorded child, mother's cognitive ability, and father's cognitive ability. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Robust standard errors in parentheses.

Table A4: Parental and child vocalizations

DV:	Child Vocalizations per Day	
	(1)	(2)
Adult female vocalizations per day	0.529***	
	(0.060)	
Adult male vocalizations per day		0.305***
		(0.109)
Observations	377	377
R^2	0.415	0.191
# Clusters	196	196
Mean DV	2480.08	2480.08
SD DV	970.43	970.43

* p< 0.10, ** p<0.05, *** p < 0.01. OLS estimates. Unit of observation is a child recording. All estimates control for the log of child's age and gender, number of children in the household, whether the mother's partner is the biological father of the child, the asset index, and a province fixed effect. Column (1) controls for mother's cognitive ability. Column (2) controls for father's cognitive ability. Missing observations imputed for cognitive ability and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

Table A5: Pitch during conversational turns

DV:	Adult female pitch		Adult male pitch	
	(1)	(2)	(3)	(4)
Conversational turn	1.951***	1.917***	1.092***	1.012***
	(0.065)	(0.079)	(0.089)	(0.113)
Conversational turn × Matriliny		0.114		0.270
		(0.145)		(0.225)
Observations	750	750	747	747
R^2	0.832	0.833	0.768	0.769
# Clusters	196	196	196	196
Mean DV	27.68	27.68	22.22	22.22
SD DV	1.52	1.52	1.55	1.55

* p< 0.10, ** p<0.05, *** p < 0.01. OLS estimates. Unit of observation is a segment of a child-parent recording. We extract 2 segments per child-parent: one when the child is vocalizing and one when the child is not. The dependent variable is the median pitch of the adult speaker's voice in the segment. Pitch is missing if no conversational turn occurs in the whole recording. Conversational turn takes the value one for the segment in which the child is vocalizing. All estimates control for a recorded child fixed effect. All estimates weighted by the duration of the recording. Standard errors clustered by child.

Table A6: Between and within village variation in beliefs

	(1)	(2)	(3)	(4)
	Mother's Returns	Mother's Costs	Father's Returns	Father's Costs
Between villages	1.91	2.00	1.98	1.90
Within villages	2.56	3.23	3.06	3.39

Table reports the between (top) and within (bottom) village standard deviations in beliefs.

Table A7: Robustness: Clustering by household or village, dropping second recorded child

DV: Adult male speech	(1)	(2)	(3)
Panel A: Adult male vocalizations			
Matriliny × Mother's Returns	101.509**	101.509***	115.350**
	(48.942)	(33.442)	(48.179)
Observations	377	377	350
Mean DV	1312.15	1312.15	1319.39
SD DV	584.47	584.47	587.53
Panel B: Adult male vocalizations at high pitch			
Matriliny × Mother's Returns	35.369*	35.369**	41.960**
	(20.085)	(14.432)	(19.264)
Observations	377	377	350
Mean DV	583.84	583.84	586.74
SD DV	272.60	272.60	273.57
Panel C: Adult male pitch during conversational turns			
Matriliny × Mother's Returns	0.260**	0.260**	0.248**
	(0.109)	(0.120)	(0.102)
Observations	371	371	345
Mean DV	22.76	22.76	22.68
SD DV	1.47	1.47	1.42
# Clusters	182	41	182
Clustering:	Household	Village	Child
Sample:	Full	Full	One Child

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. Unit of observation is a child recording. Each panel heading A-C indicates the dependent variable. All estimates control for father's returns and costs, each interacted with matriliny, as well as the log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, father's cognitive ability, the asset index, and a province fixed effect. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the house level in column (1), village level in column (2), and child level in column (3). Column (3) is additionally restricted to one child recording per household (keeping only the one closest to 24 months or, in the case of twins, a child drawn randomly). All estimates weighted by the duration of the recording.

Table A8: Father characteristics by mother's returns and matriliny

DV:	Returns of CDS (1)	Costs of CDS (2)	Age (3)	Secondary School (4)	Income: Agric (5)	Income: Fishing (6)	Leisure Time (7)	Trust (8)	Reciprocity (9)	Care from Mother (10)	Care from MiL (11)	Care from Friends (12)
Mother's Returns × Matriliny	0.172	-0.192	-0.488	0.015	-0.022	0.049	-0.068	0.099	-0.044	0.007	0.090	0.043
	(0.322)	(0.351)	(0.592)	(0.044)	(0.039)	(0.043)	(0.116)	(0.108)	(0.248)	(0.093)	(0.091)	(0.092)
Matriliny	-1.100	1.556	0.468	0.094	0.169	-0.302	0.124	-0.436	-0.763	-0.162	0.150	-0.058
	(1.543)	(1.681)	(2.834)	(0.212)	(0.188)	(0.205)	(0.510)	(0.519)	(1.186)	(0.446)	(0.438)	(0.439)
Mother's Returns	-0.045	-0.099	0.127	0.009	0.003	-0.010	0.047	-0.023	0.126	-0.012	-0.004	0.002
	(0.137)	(0.149)	(0.253)	(0.019)	(0.017)	(0.018)	(0.045)	(0.046)	(0.105)	(0.040)	(0.039)	(0.039)
Observations	157	158	157	158	158	158	121	158	158	158	158	158
R^2	0.066	0.048	0.049	0.032	0.061	0.045	0.038	0.055	0.081	0.015	0.051	0.047
Mean DV	4.15	4.41	33.40	0.35	0.25	0.32	-0.00	2.47	7.67	-0.00	-0.00	0.00
SD DV	3.55	3.83	6.47	0.48	0.43	0.47	1.00	1.19	2.75	1.00	1.00	1.00

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. OLS estimates. Unit of observation is the father of a recorded child. Each column is the result of regressing the dependent variable indicated at the top of the column on mother's returns, matriliny, and the interaction, controlling for province fixed effects. *Returns* and *costs* of CDS are our usual measures. *Leisure* is the time spent on leisure, standardized to mean zero, standard deviation one. *Trust* measured as a response to the following question: "Generally speaking, would you say that most people can be trusted, or that you can't be too careful in dealing with people?", where 1=complete distrust and 5=complete trust. *Reciprocity* measured as a response to the following question: "How well does the following statement describe you as a person? I try especially hard to help someone who has helped me in the past, where 0=not at all and 10=describes me perfectly". *Care from* is the amount of help with child care from the parent's own mother, mother-in-law (MiL), and friends and neighbors, each measured as: 0=never, 1=less than once a month, 2=more than once a month, 3=once a week, 4=couple times a week, 5=every day, a little bit, 6=every day, for a large part of the day; values standardized to mean zero, standard deviation one. Missing observations imputed for beliefs and all estimates include indicator variables for missing observations.

Table A9: Child, household, and mother characteristics by mother's returns and matriliny

DV:	Child BMI (1)	Child Female (2)	Mother's Age (3)	Spouse Not Biofather (4)	Children in HH (5)	Secondary School (6)	Income: Agric (7)	Income: Fishing (8)	Leisure Time (9)	Trust (10)	Reciprocity (11)	Care from Mother (12)	Care from MiL (13)	Care from Friends (14)
Mother's Returns × Matriliny	-0.082	-0.011	-0.101	-0.009	0.023	0.001	0.014	0.019	0.080	0.044	-0.124	0.088	-0.021	0.084
	(0.124)	(0.041)	(0.475)	(0.031)	(0.138)	(0.039)	(0.042)	(0.031)	(0.109)	(0.087)	(0.197)	(0.084)	(0.083)	(0.084)
Matriliny	0.294	0.112	-1.759	0.096	-0.694	0.171	-0.182	0.083	-0.406	-0.111	1.556*	-0.082	-0.042	-0.111
	(0.591)	(0.196)	(2.280)	(0.147)	(0.659)	(0.184)	(0.199)	(0.149)	(0.502)	(0.413)	(0.940)	(0.401)	(0.398)	(0.399)
Mother's Returns	0.051	-0.039**	0.092	-0.007	0.014	-0.006	-0.006	-0.002	-0.017	-0.060	0.143*	-0.030	-0.030	-0.016
	(0.054)	(0.018)	(0.209)	(0.013)	(0.060)	(0.017)	(0.018)	(0.014)	(0.044)	(0.038)	(0.086)	(0.037)	(0.036)	(0.036)
Observations	176	182	181	182	182	182	182	182	115	182	182	182	182	182
R^2	0.049	0.060	0.033	0.042	0.035	0.027	0.034	0.041	0.075	0.061	0.129	0.016	0.028	0.026
Mean DV	15.23	0.46	29.38	0.16	3.03	0.31	0.48	0.17	0.00	2.53	7.41	-0.00	0.00	-0.00
SD DV	1.49	0.50	5.69	0.37	1.66	0.46	0.50	0.38	1.00	1.05	2.49	1.00	1.00	1.00

* p< 0.10, ** p<0.05, *** p < 0.01. OLS estimates. Unit of observation is the mother of a recorded child. Each column is the result of regressing the dependent variable indicated at the top of the column on mother's returns, matriliny, and the interaction, controlling for province fixed effects. *Leisure* is the time spent on leisure, standardized to mean zero, standard deviation one. *Trust* measured as a response to the following question: "Generally speaking, would you say that most people can be trusted, or that you can't be too careful in dealing with people?", where 1=complete distrust and 5=complete trust. *Reciprocity* measured as a response to the following question: "How well does the following statement describe you as a person? I try especially hard to help someone who has helped me in the past, where 0=not at all and 10=describes me perfectly". *Care from* is the amount of help with child care from the parent's own mother, mother-in-law (MiL), and friends and neighbors, each measured as: 0=never, 1=less than once a month, 2=more than once a month, 3=once a week, 4=couple times a week, 5=every day, a little bit, 6=every day, for a large part of the day; values standardized to mean zero, standard deviation one. Missing observations imputed for beliefs and all estimates include indicator variables for missing observations.

Table A10: Main results: conditioning on child care by father

DV:	Vocs	Vocs at High Pitch	Pitch
Speaker: Adult male	(1)	(2)	(3)
Matriliny × Mother's Returns	98.550**	33.770*	0.273***
	(48.766)	(19.824)	(0.102)
Matriliny	71.280	100.694	-1.263*
	(381.603)	(170.379)	(0.684)
Mother's Returns	-19.450	-5.716	-0.072*
	(19.017)	(7.987)	(0.043)
Father provides daily child care	32.464	20.654	-0.194
	(81.364)	(36.033)	(0.200)
Observations	377	377	371
R^2	0.18	0.22	0.23
# Clusters	196.00	196.00	196.00
Mean DV	1312.15	583.84	22.76
SD DV	584.47	272.60	1.47

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. Unit of observation is a child recording. The dependent variable in column (1) is the number of adult male vocalizations per day. The dependent variable in column (2) is the number of adult male vocalizations at high pitch per day, where high pitch is above the speaker's average median pitch during conversational turns with the child. The dependent variable in column (3) is the average median pitch of the adult male speaker's voice during conversational turns with the child. Pitch is missing if no conversational turn occurred during the recording. All estimates control for a binary variable equal to one if the father takes care of the child on a daily basis, the log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, father's cognitive ability, the asset index, and a province fixed effect. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

Table A11: Village characteristics as potential sources of female bargaining power

Village Characteristic:	Market in Vill	Kindergarten	Primary School	Elected Leader
DV: Adult male speech	(1)	(2)	(3)	(4)
Panel A: Adult male vocalizations				
Matriliny × Mother's Returns	98.479**	98.192**	114.004**	103.208**
	(49.505)	(48.229)	(46.973)	(45.665)
Observations	377	377	377	377
Mean DV	1312.15	1312.15	1312.15	1312.15
SD DV	584.47	584.47	584.47	584.47
Panel B: Adult male vocalizations at high pitch				
Matriliny × Mother's Returns	33.594*	33.596*	39.935**	31.594*
	(19.926)	(19.540)	(18.346)	(19.156)
Observations	377	377	377	377
Mean DV	583.84	583.84	583.84	583.84
SD DV	272.60	272.60	272.60	272.60
Panel C: Adult male pitch during conversational turns				
Matriliny × Mother's Returns	0.288***	0.265**	0.263**	0.301***
	(0.102)	(0.111)	(0.104)	(0.107)
Observations	371	371	371	371
Mean DV	22.76	22.76	22.76	22.76
SD DV	1.47	1.47	1.47	1.47
# Clusters	196	196	196	196
Village characteristic	Y	Y	Y	Y
Village characteristic x beliefs	Y	Y	Y	Y
Mean village characteristic	0.44	0.68	0.68	0.34

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. Unit of observation is a child recording. Each specification regresses a measure of adult male speech (indicated in each panel heading A-C) on matriliny and its interaction with parental beliefs, conditioning on a village characteristic, indicated a the top of the column, interacted with parental beliefs. Column (1) controls for whether there is a market in the village. Column (2) controls for whether there is a kindergarten in the village. Column (3) controls for whether there is a primary school in the village. Column (4) controls for whether an elected leader governs the village. All estimates control for father's returns and costs interacted with *matriliny*, as well as the log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, father's cognitive ability, the asset index, and a province fixed effect. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

Table A12: Child care and heterogeneity by child gender - Pure matriliny

DV:	Father Child Care		Adult Male Speech		
			Vocs	Vocs at High Pitch	Pitch
	(1)	(2)	(3)	(4)	(5)
Full Matri	0.417*	0.834	898.547**	425.986***	-1.179
	(0.220)	(0.613)	(375.057)	(161.053)	(0.825)
Full Matri × Mother's Returns		-0.031	-0.284	-3.047	0.346**
		(0.083)	(44.931)	(19.913)	(0.134)
Female child × Full Matri			-1446.969***	-562.621***	-0.690
			(329.322)	(143.599)	(0.801)
Female child × Full Matri × Mother's Returns			213.471***	82.679***	0.088
			(61.571)	(26.865)	(0.171)
Observations	196	196	377	377	371
R^2	0.26	0.28	0.29	0.32	0.27
Mean DV	0.00	0.00	1,312.15	583.84	22.76
SD DV	1.00	1.00	584.47	272.60	1.47

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. *Full Matri* is a dummy variable taking value one in fully matrilineal villages (mean: 0.11, s.d.: 0.31, see Table A1). In columns (1)-(2), the unit of observation is a household, and the dependent variable is the how often the father takes care of the children, standardized to mean zero, standard deviation one. In columns (3)-(5), the unit of observation is a child recording. In column (3), the dependent variable is adult male vocalizations per day. In column (4), the dependent variable is the number of adult male vocalizations at high pitch per day, where high pitch is above the speaker's average median pitch during conversational turns with the child. In column (5), the dependent variable the average median pitch of the adult male speaker's voice during conversational turns with the child. Pitch is missing if no conversational turn occurred during the recording. All estimates control for log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, the asset index, father's cognitive ability, and a province fixed effect. Columns (3)-(5) also control for father's returns and costs, interacted with *Full Matri*. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Robust standard errors in parentheses in columns (1)-(2). Columns (3)-(5) cluster standard errors at the child level and weight estimates by the duration of the recording.

Table A13: Returns, costs, and vocalizations - Mothers: coefficients associated with cost characteristics

DV: Adult female vocalizations per day	(1)	(2)	(3)	(4)
Mother's Returns	36.171*	38.542*	38.370*	55.555
	(21.929)	(22.806)	(22.781)	(41.863)
Mother's Costs	16.798	16.240	13.526	23.327
	(16.374)	(16.437)	(16.329)	(28.189)
Father's Returns		-12.512	6.093	0.038
		(21.858)	(36.936)	(45.594)
1/Matriliny × Father's Returns			-2.921	-1.985
			(4.783)	(6.102)
1/Matriliny × Mother's Returns				-2.630
				(5.580)
1/Matriliny × Mother's Costs				-1.420
				(3.803)
1/Matriliny	28.014	28.426	41.864	54.533
	(17.291)	(17.477)	(30.132)	(52.206)
log(child's age)	-446.603***	-449.478***	-452.289***	-460.353***
	(147.139)	(148.047)	(148.785)	(146.894)
Female child	-203.197*	-207.269*	-204.272*	-207.432*
	(120.329)	(122.636)	(123.722)	(125.101)
# Children in HH	29.761	29.061	27.979	29.001
	(38.644)	(38.408)	(38.681)	(38.799)
Spouse not biological father	-51.658	-48.914	-51.846	-47.170
	(179.330)	(179.154)	(178.952)	(184.919)
Mother's cognitive ability	11.731	11.532	11.007	10.944
	(7.879)	(7.866)	(7.986)	(8.005)
Asset index	74.712**	75.390**	75.344**	74.891**
	(36.919)	(37.010)	(37.034)	(36.792)
R^2	0.23	0.23	0.23	0.22
Mean DV	2,782.89	2,782.89	2,782.89	2,782.89
SD DV	988.35	988.35	988.35	988.35

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays mothers' output only. Unit of observation is a child recording. All estimates include a province fixed effect. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

Table A14: Returns, costs, and vocalizations - Fathers: coefficients associated with cost characteristics

DV: Adule male vocalizations per day	(1)	(2)	(3)	(4)
Father's Returns	-4.256	-7.865	-7.617	-3.203
	(12.842)	(14.521)	(14.868)	(17.574)
Father's Cost	-18.369*	-18.047*	-16.244*	-13.825
	(9.932)	(9.874)	(9.755)	(12.283)
Mother's Returns		3.699	-18.437	-20.191
		(16.421)	(18.793)	(18.816)
Matriliny × Mother's Returns			95.212**	101.509**
			(46.040)	(47.820)
Matriliny × Father's Returns				-15.558
				(52.053)
Matriliny × Father's Cost				-9.177
				(25.635)
Matriliny	354.039***	354.756***	-6.307	81.629
	(125.169)	(126.588)	(213.645)	(381.874)
log(child's age)	-29.715	-31.336	-47.640	-42.137
	(103.490)	(103.194)	(102.283)	(99.431)
Female child	-90.397	-87.472	-80.220	-84.221
	(79.089)	(82.847)	(83.336)	(83.591)
# Children in HH	-8.755	-9.254	-8.912	-7.551
	(25.596)	(25.518)	(25.596)	(25.524)
Spouse not biological father	-102.713	-100.111	-86.697	-88.202
	(100.545)	(100.481)	(100.568)	(100.774)
Father's cognitive ability	0.827	0.572	0.962	0.883
	(5.609)	(5.764)	(5.803)	(5.787)
Asset index	-15.763	-15.782	-16.509	-17.545
	(25.482)	(25.314)	(24.210)	(24.315)
R^2	0.15	0.15	0.18	0.18
Mean DV	1,312.15	1,312.15	1,312.15	1,312.15
SD DV	584.47	584.47	584.47	584.47

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays fathers' output only. Unit of observation is a child recording. All estimates include a province fixed effect. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

B Additional Robustness

In this section, we establish the robustness of our results to alternative modeling choices for the beliefs, to different functional form assumptions for costs and for the relationship between matrilineal inheritance and female bargaining power, and to alternative hypotheses about parental utility functions.

B.1 Expectations about the returns to parental investment and cost of Investment

B.1.1 Data quality checks on the probabilistic expectations

We measure probabilistic expectations using a visual aid as is commonly done in low-income countries (Delavande, 2023). We start with a preamble explaining the notion of a probability (see Appendix E for exact wording). We then have a practice question where we elicit the likelihood of a woman in the community going to the market (a) in the next 2 days and (b) in the next 2 weeks. The majority of respondents recognize that the probability of going to the market is higher the longer the time span allowed. However, about 20% of respondents violate monotonicity, and this is similar by gender and matriliny. This percentage is slightly higher than found in similar aged-sample in other low-income countries (Delavande and Kohler, 2009; Bhalotra et al., 2024), and on par with a sample of older respondents in India (Delavande et al., 2017).

We then ask parental beliefs about the likelihood that a child puts 2-3 words together in speaking by age 2 if the carers frequently (vs. rarely) talk to the child directly. These questions are framed with a reference to a typical carer and child. The advantage of using this wording rather than the actual carer-child pair is that the latter may be correlated with unobserved carer or child characteristics, which could bias our estimates. For this reason, our approach is the standard in the literature (Cunha et al., 2013; Attanasio et al., 2019; Bhalotra et al., 2024).

Respondents are willing to report their beliefs in a probabilistic manner. Non-response rates on the probabilistic and cost questions are low for mothers (9%). It is slightly higher for fathers (14% for returns, 13% for costs), which is primarily explained by the father being absent at the time of interview.

Our main results are based on using the expected return of child-directed speech, which is the individual-level difference in the probability of reaching the language milestone between the high and low investment. Eight percent of men and 5% of women report a negative return, which may be an indication of lack of attention or difficulty with the question.

Table B1 presents the conditional association of expected returns and cost separately for mothers and fathers with basic characteristics. Average returns and costs do not vary with matriliny, neither for mothers, nor for fathers. Mothers expect lower returns for girls and report higher costs if they have more children. Men who score higher on a cognitive task expect higher returns. Mothers who are married to a man who is not the child's biological father also report lower cost of investment. Assets have no predictive power on returns or costs.

Table B1: Descriptive correlates of expected returns and cost to child-directed speech

DV: Parental beliefs	Mothers		Fathers	
	Returns	Costs	Returns	Costs
	(1)	(2)	(3)	(4)
Matriliny	-0.520	0.045	-0.574	0.539
	(0.613)	(0.847)	(0.624)	(0.880)
Female child	-1.192***	-0.216	-0.548	-0.015
	(0.400)	(0.555)	(0.411)	(0.579)
ln(age)	-0.102	0.927	-0.363	0.114
	(0.499)	(0.691)	(0.498)	(0.704)
# Children in HH	0.102	0.269	-0.045	-0.057
	(0.121)	(0.168)	(0.122)	(0.172)
Spouse not biofather	-0.732	-1.281*	0.092	0.334
	(0.532)	(0.737)	(0.539)	(0.764)
Mother's cognitive ability	0.014	-0.013		
	(0.031)	(0.043)		
Father's cognitive ability			0.065*	0.006
			(0.034)	(0.048)
Asset index	-0.013	0.187	0.071	-0.065
	(0.110)	(0.152)	(0.117)	(0.166)
Observations	182	182	182	182
R^2	0.215	0.071	0.266	0.017
Mean DV	4.33	4.88	4.32	4.41
SD DV	2.80	3.58	2.91	3.57

* p< 0.10, ** p<0.05, *** p < 0.01. OLS estimates. Unit of observation is the parent of a recorded child. All estimates control for a province fixed effect. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations.

B.1.2 Robustness to measurement error in beliefs

In our main specification, we replace missing returns and costs with their sample average and include a dummy variable for missing. We also replace negative returns with zero. We keep all respondents in our estimation sample, even those who violated the monotonicity property of probabilities in the preamble questions. We now investigate the robustness of our main result, that males vocalize more in matrilineal villages when the mother expects

higher returns, to these modeling decisions.

Our results for quantity of vocalizations and pitch are, if anything, larger in magnitude when we exclude monotonicity violators (Column 1 of Table B2). Our results are also robust to keeping the negative returns as negative (Column 2 of Table B2) and to excluding respondents with missing returns (Column 3 of Table B2).

One potential limitation of our measure of expected returns is that it may be measured with error, because it is difficult to precisely specify the parental investment and the child outcome in the different scenarios. We address this possible concern by taking instead an average across ours and several other measures of returns to speech in our survey (as in Cunha et al. (2020); Bhalotra et al. (2024)). Specifically, our survey also captures parental beliefs about the returns to speech for (i) child school readiness by the age of 6, and (ii) child educational performance in primary school (see Appendix E). These measure outcomes when the child is of primary school age and are asked in relation to the actual parent-child pair rather than a hypothetical pair, which is why they are not our preferred measures of return. We average our main return for language development with these two additional measures. Our findings with this alternative measure (Column 4 of Table B2) align closely with the main results in Tables 3 and 4.

Table B2: Dropping violators of monotonicity, including negative returns, dropping missing observations, and using average returns

DV: Adult male speech	(1)	(2)	(3)	(4)
Panel A: Adult male vocalizations				
Matriliny × Mother's Returns	101.509**	94.366**	78.167*	88.743*
	(47.820)	(42.623)	(46.472)	(49.173)
Observations	377	377	296	377
Mean DV	1312.15	1312.15	1345.97	1312.15
SD DV	584.47	584.47	601.57	584.47
Panel B: Adult male vocalizations at high pitch				
Matriliny × Mother's Returns	35.369*	32.737*	24.080	33.127
	(19.356)	(16.940)	(17.998)	(20.200)
Observations	377	377	296	377
Mean DV	583.84	583.84	598.63	583.84
SD DV	272.60	272.60	277.42	272.60
Panel C: Adult male pitch during conversational turns				
Matriliny × Mother's Returns	0.260**	0.217**	0.262***	0.253**
	(0.102)	(0.094)	(0.101)	(0.111)
Observations	371	371	290	371
Mean DV	22.76	22.76	22.65	22.76
SD DV	1.47	1.47	1.47	1.47
# Custers	196	196	196	196
Drop violators of monotonicity:	Y	N	N	N
Include negative returns:	N	Y	N	N
Drop missing observations:	N	N	Y	N
Use average returns:	N	N	N	Y

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. Unit of observation is a child recording. Each panel heading A-C indicates the dependent variable. Column (1) drops violators of monotonicity. Column (2) includes negative returns. Column (3) drops observations with missing returns and costs information. Column (4) uses the average of returns for speech, school readiness, and primary school performance. All estimates control for fathers returns and costs, each interacted with matriliny, as well as the log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, the asset index, father's cognitive ability, and a province fixed effect. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

B.2 Robustness to alternative theoretical assumptions

B.2.1 Functional form assumptions of the cost function and bargaining power shifter

We explore different assumptions with implications for the estimation procedure. From this point forward, we normalize the cost function by setting $\chi_{i,2} = 1$ for $i = m, f$. Under this normalization, the optimal investments strategies x_m^* and x_f^* are given by:

$$x_m^* = \omega_m \eta_{m,2} + (\frac{1-\alpha}{\alpha})\omega_f \eta_{f,2} - \chi_{m,1} \tag{14}$$

$$x_f^* = \omega_f \eta_{f,2} + (\frac{\alpha}{1-\alpha})\omega_m \eta_{m,2} - \chi_{f,1} \tag{15}$$

With the normalization, the preference parameters ω_m and ω_f are both identified. We estimate Equations (14) and (15) under two different assumptions of the relationship between the mother's pareto weight and matriliny.

Linear Relationship. We first maintain the linearity assumption and further assume that $\frac{\alpha}{1-\alpha} = M$. Under this assumption, the preference parameters ω_m and ω_f are both identified from both equations. We estimate them jointly using maximum likelihood with the following equations:

$$\begin{aligned} Speech_{m,c,h} = a_m + \boldsymbol{\delta_1} R_m + \delta_{2,m} * R_f + \boldsymbol{\delta_3} \frac{1}{M_h} * R_f \\ + \kappa_{1,m} C_m + \gamma_m X_{m,c,h} + \epsilon_{m,c,h} \end{aligned} \tag{16}$$

$$\begin{aligned} Speech_{f,c,h} = a_f + \boldsymbol{\delta_3} R_f + \delta_{2,f} * R_m + \boldsymbol{\delta_1} M_h * R_f \\ + \kappa_{1,f} C_f + \gamma_m X_{f,c,h} + \epsilon_{f,c,h}, \end{aligned} \tag{17}$$

Note that δ_1, which estimates the structural parameter ω_m, and δ_3, which estimates the structural parameter ω_f, are identified through the joint estimation of both parental investment equations. We are primarily interested in $\delta_1 which measures the elasticity of father's investment to moti$

The estimation results are presented in Table B3. Our key finding is that fathers' investments are responsive to mothers' beliefs when mothers have greater bargaining power. Specifically, the results indicate that δ_1 is positive and statistically significant across all measures of paternal investments, including conversational turns. This highlights the robustness of our findings to different assumptions and estimation procedures.

Logistic Relationship. Second, we relax the assumption of linearity of $\frac{\alpha}{1-\alpha}$ in matriliny. We consider instead that the mother's pareto weight is a logistic function of matriliny, so that:

$$\alpha(M) = \frac{1}{1 + \exp -(\beta_0 + \beta_1 M)} \tag{18}$$

Under this assumption, the preference parameters ω_m and ω_f, as well as the parameters β_0 and β_1, are identified and are included in both equations. As before, we estimate the

equations jointly. The equations for estimations are similar to Equations (16) and (17), with the key difference that M_h takes the functional form specified by Equation (18). We employ feasible generalized nonlinear least squares for estimation. A subset of estimated parameters are displayed in Tables B4. We find a positive and precisely estimated δ_1, suggesting that mothers' returns are a positive driver of paternal investments when she has higher bargaining power. The results fall short of statistical significance for pitch but are statistically significant for all quantitative measures of investments: total vocalizations, vocalizations at high pitch, as well as conversational turns.

Table B3: Robustness to joint estimation and different assumptions on cost parameters

DV:	Vocs	CTC	Vocs at High Pitch	Pitch
Speaker: Adult female and male	(1)	(2)	(3)	(4)
Mother's return (δ_1)	49.112**	19.362*	23.322***	0.222**
	(20.965)	(11.623)	(8.822)	(0.087)
Father's return (δ_3)	-3.094	0.346	-1.617	0.049
	(4.564)	(2.796)	(1.978)	(0.035)
Mother's cost ($\kappa_{1,m}$)	12.917	11.757	7.468	-0.023*
	(16.280)	(13.397)	(7.135)	(0.014)
Father's cost ($\kappa_{1,f}$)	-15.782	-6.424	-6.167	0.041*
	(9.720)	(5.109)	(4.315)	(0.024)
Observations	377	377	377	371
# Clusters	196	196	196	196
R^2	0.16	0.25	0.21	0.22
Mean DV	1,312.15	441.93	583.84	22.76
SD DV	584.47	271.56	272.60	1.47

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations. The parameters $\delta_1 and \delta_3 are estimated simultaneously from both equations. Unit of observation is a child recording. The dependent variables are:$ $the number of adult male vocalizations per day (column 1), the number of conversational turns per day between the child and adult speaker (column 2), th$ $the log of child' s age, child' s gender, number of children in the household, whether the mother' s partner is the biological father of the child, father' s co$

Table B4: Robustness to joint estimation and non-linear assumptions on Pareto weights

DV:	Vocs	CTC	Vocs at High Pitch	Pitch
Speaker: Adult female and male	(1)	(2)	(3)	(4)
Mother's returns (δ_1)	43.638*	42.365**	21.313*	0.020
	(26.241)	(19.676)	(11.798)	(0.018)
Father's returns (δ_3)	0.000	0.000	0.000	0.001
	(0.000)	(0.000)	(0.000)	(0.010)
Mother's cost ($\kappa_{1,m}$)	6.833	10.322	4.283	-0.018
	(18.138)	(14.094)	(8.021)	(0.014)
Father's cost ($\kappa_{1,f}$)	-15.925	-5.974	-5.802	0.037
	(9.802)	(4.900)	(4.378)	(0.023)
Pareto Weight				
β_0	-17.700	-17.247	-17.034	-2.833
	(35.053)	(42.557)	(33.335)	(7.787)
β_1 (Matriliny)	20.654	18.617	19.809	4.730
	(38.948)	(47.317)	(37.043)	(7.854)
Observations	377	377	377	371
R^2 Mother	0.105	0.203	0.143	0.120
R^2 Father	0.155	0.215	0.199	0.158

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the non-linear joint estimation of mother and father equations. The parameters

$\delta_1 and \delta_3 are estimated simultaneously from both equations. Unit of observation is a child recording. All estimates control for the lo$

B.2.2 Alternative cost functions

We evaluate the robustness of our results to the inclusion of various characteristics into the cost functions. As discussed in Section 6, Table B5 demonstrates that our findings remain consistent across a range of alternative assumptions regarding the structure of the cost function.

Table B5: Robustness: Alternative cost function

DV: Adult male speech	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Adult male vocalizations									
Matriliny × Mother's Returns	106.836**	106.991**	101.509**	94.983*	85.559*	98.841**	114.588**	107.354**	103.164**
	(46.865)	(47.720)	(47.820)	(50.897)	(45.223)	(44.765)	(49.091)	(44.917)	(48.312)
Observations	377	377	377	377	377	377	377	377	377
Mean DV	1312.15	1312.15	1312.15	1312.15	1312.15	1312.15	1312.15	1312.15	1312.15
SD DV	584.47	584.47	584.47	584.47	584.47	584.47	584.47	584.47	584.47
Panel B: Adult male vocalizations at high pitch									
Matriliny × Mother's Returns	38.743**	37.841*	35.369*	33.654*	29.163	30.116*	37.276*	34.469*	36.897*
	(19.408)	(19.543)	(19.356)	(20.154)	(18.472)	(17.389)	(19.298)	(18.503)	(19.704)
Observations	377	377	377	377	377	377	377	377	377
Mean DV	583.84	583.84	583.84	583.84	583.84	583.84	583.84	583.84	583.84
SD DV	272.60	272.60	272.60	272.60	272.60	272.60	272.60	272.60	272.60
Panel C: Adult male pitch during conversational turns									
Matriliny × Mother's Returns	0.236**	0.250**	0.260**	0.244**	0.268***	0.287***	0.160*	0.285***	0.264***
	(0.102)	(0.102)	(0.102)	(0.096)	(0.103)	(0.099)	(0.094)	(0.107)	(0.100)
Observations	371	371	371	371	371	371	371	371	371
Mean DV	22.76	22.76	22.76	22.76	22.76	22.76	22.76	22.76	22.76
SD DV	1.47	1.47	1.47	1.47	1.47	1.47	1.47	1.47	1.47
# Clusters	196	196	196	196	196	196	196	196	196
Controls:	None	Child	Full	All interactions	Full+Care	Full+Language	Full+Date	Full+Income	Full+Child Vocs

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and displays father's results only. Unit of observation is a child recording. Each panel heading A-C indicates the dependent variable. Column (1) shows our main result in an uncontrolled regression. Column (2) adds only the following child controls: the log of child's age and child's gender, as well as a dummy variable for the household's randomized treatment assignment in the child-directed speech intervention baseline. Columns (3)-(6) add the following controls: the number of children in the household, whether the mother's partner is the biological father of the child, the asset index, and father's cognitive ability. Column (3) thus displays our baseline coefficient. Column (4) interacts all controls with matriliny. Column (5) additionally controls for the total amount of help with child care that the mother receives from female and male caregivers, separately, in addition to the full controls. Column (6) includes language fixed effects. Column (7) controls for date of recording fixed effects. Column (8) controls for father's income source. Column (9) controls for total vocalizations from children who are not the key child. In all specifications, standard errors are clustered at the child level. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. All estimates weighted by the duration of the recording.

B.2.3 Spousal altruism

In the utility function we posit in Equation 1, we assume that each parent derives utility about the child's human capital but does not internalize the cost of parental investment born by their spouse (the other parent). This assumption would be violated if spouses are altruistic and care about each other's costs. Moreover, a literature on matrilineal inheritance (e.g. Lowes (2022)) suggests that matrilineal inheritance may reduce spousal altruism by creating split familial allegiances between spouses, thereby implying heterogeneity in spousal altruism by inheritance structure.[45]

To account for spousal altruism, we add the spouse's direct cost into the parent's cost function. We therefore estimate the optimal joint parental investment as follows:

$$\begin{aligned} Speech_{m,c,h} = a_m + \delta_{1,m} R_m + \delta_{2,m} * R_f + \delta_{3,m} \frac{1}{M_h} * R_f + \delta_{4,m} \frac{1}{M_h} * C_f \\ + \kappa_{1,m} C_m + \gamma_m X_{m,c,h} + \epsilon_{m,c,h} \end{aligned} \tag{19}$$

$$\begin{aligned} Speech_{f,c,h} = a_f + \delta_{1,f} R_f + \delta_{2,f} * R_m + \delta_{3,f} M_h * R_m + \delta_{4,f} M_h * C_m \\ + \kappa_{1,f} C_f + \gamma_m X_{f,c,h} + \epsilon_{f,c,h}, \end{aligned} \tag{20}$$

Table B6 displays the estimation results for our three main measures of paternal investment. Our main results are robust: fathers whose spouse expects higher returns invest more in verbal engagement with their children in matrilineal villages, across all quantitative and qualitative measures. The estimated coefficients and standard errors hardly change when the cross direct cost is included. The coefficient associated with spousal cost is never statistically significant at conventional levels, either on its own or interacted with matriliny, suggesting that potential heterogeneity in spousal altruism across inheritance structures is not relevant in our context.

[45] Split allegiances emerge when a husband also has allegiances to his sisters, whose children he may support because they are his heirs; and the wife has allegiances to her brothers, who provide her and her children with support. Such split allegiances are particularly relevant in avunculocal matrilineal systems, where children inherit from their maternal uncles, which is not the form of inheritance practiced in our setting (see Footnote 12).

Table B6: Robustness to spousal altruism

DV: Adult male speech	Vocs	Vocs at High Pitch	Pitch
	(1)	(2)	(3)
Matriliny × Mother's Returns	96.389**	33.431*	0.288***
	(47.378)	(19.451)	(0.102)
Matriliny × Mother's Costs	-55.735*	-13.538	0.024
	(33.704)	(15.891)	(0.079)
Matriliny	219.737	142.325	-1.553**
	(409.071)	(186.232)	(0.776)
Mother's Returns	-20.689	-6.105	-0.072*
	(18.626)	(7.775)	(0.042)
Mother's Costs	10.055	0.970	0.039
	(14.108)	(6.481)	(0.035)
Father's Returns	-4.257	-3.684	0.052
	(17.606)	(8.124)	(0.042)
Father's Cost	-22.737*	-8.045	0.015
	(13.647)	(6.263)	(0.034)
Observations	377	377	371
R^2	0.18	0.22	0.24
# Clusters	196	196	196
Mean DV	1,312.15	583.84	22.76
SD DV	584.47	272.60	1.47

* p< 0.10, ** p<0.05, *** p < 0.01. Table presents results from the joint estimation of mother and father equations, and presents father results, only. Unit of observation is a child recording. All estimates control for the interaction of father's returns and costs with matriliny, the log of child's age, child's gender, number of children in the household, whether the mother's partner is the biological father of the child, the asset index, father's cognitive ability, and a province fixed effect. Missing observations imputed for beliefs and cognitive ability, and all estimates include indicator variables for missing observations. Standard errors clustered at the child level in parentheses. All estimates weighted by the duration of the recording.

C Details on the measurement of speech

We measure speech from child-centered long-form recordings collected using a wearable device. In urban conditions, with reliable access to electricity and the internet, a hardware-software combined solution called LENA is frequently used (Gilkerson et al., 2017). LENA is suboptimal for the current approach for two key reasons. First, the lack of electricity and internet connection found in the remote communities of the Solomon Islands imply that the licenses promoted by LENA are inappropriate: to recover the audio-recordings, one needs to connect the recording device to a computer, have enough power and internet to extract and upload the recording (which is meanwhile stored in a proprietary format) before the recording device can be used again. Second, LENA's software was trained in urban conditions among monolingual American English learners, which raised questions about whether the software would be sufficiently accurate in our population of largely multilingual infants (children in our sample are exposed to one or more of 12 different languages: Avaso, Babatana, Marco, Marovo, Pidjin, Roviana, Senga, Simbo, Sisinga, Ughele, Vaghua, Varisi) growing up in rural conditions.

As we explain in Section 3.4, the Voice Type Classifier (VTC; Lavechin et al. (2020)) was trained with a combination of various child-centered corpora of children aged 0-4 years exposed to one or more of a variety of languages (including Minn, French, Jul'hoan, Tsimané, English, and several others, in approximate order of data quantity). These corpora included children growing up in multilingual settings, as well as languages spoken in the Pacific, with a wide variety of typological characteristics. The multi-corpus training was done to improve the generalizability of the network to unseen data sets. The corpora were divided internally into independent training, developmental, and testing sets. As reported in Lavechin et al. (2020), F-score performance on the test set of this multilingual corpus was 77.3% for recognizing the key child.[46] Lavechin et al. (2020) also reports on performance for a wholly independent, unseen, test set comprised on monolingual English learners, for which LENA performance was also available. In that comparative dataset, LENA's performance for the key child was 54.9%, whereas the VTC scored nearly 25% higher, at 68.7%. As mentioned in the Introduction, VTC also outperformed LENA for male (F-score of 43% versus 37%) and female voices (F-score of 63% versus 43%; Lavechin et al., 2020).

We also checked the performance of VTC in a subset of data analyzed in the present paper. About 87 minutes of audio were annotated by research assistants who were unfamiliar with the language and the families that were recorded, which may have a negative impact on the accuracy of their annotations. Indeed, human-human F-score on a subset of the same data (totalling 67 minutes) was not 100%. Since two humans can disagree with each other, we are probably underestimating the accuracy of VTC because human-human agreement provides an upper bound for human-VTC agreement. We observe the following F-scores: Key child

[46] F-score is a measure of accuracy that goes from 0 to 100%, higher is better. In technical terms, it is the harmonic mean of precision and recall, with precision being the percentage of retrieved elements that are true positives, and recall the percentage of relevant elements that were indeed retrieved. For instance, if the "key child" category has a precision of 70%, that means that 30% of the times that the system returned "key child", it was wrong, and it was a different category (e.g., female adult). A recall of 70% means that 30% of the times the key child vocalized, the system did not spot this (e.g., missed the speech altogether or put it in the "female adult" category).

61% for VTC-human, versus 83% human-human; female adult 59% for VTC-human versus 82% human-human; male adult 39% for VTC versus 79% human-human. Although there is still room for algorithm improvement before it can match human-human agreement, the relatively lower level of accuracy for an algorithm compared to humans is offset by the quantity of data that can be used to represent each individual child.

Another limitation of LENA consists of measurement error in automated measures of conversational turn counts (CTC). In a meta-analysis looking at the reliability of automated measures, Cristia et al. (2020) found that LENA's CTC had only a .35 correlation with human coded counts for conversational turns. In contrast, the measures of child or adult vocalization counts correlated highly with human estimates (r>.7), suggesting that vocalizations are more precise measure of language compared with CTC turns. To some extent, this may be explained by the fact that LENA uses a 5-second frame to code adult-child or child-adult sequence. 5 seconds may be too long, considering that a recent meta-analysis suggests most child-adult and adult-child turn transitions occur within 2 seconds (Nguyen et al., 2022). To account for this, we use a 1 second threshold in our CTC measure.[47]

D Definition of key variables

Below, we define the key variables in our analysis:

- *Matriliny*: reported by the village leader in response to the following question: "How is land inherited in this village?" – (i) father; (ii) mother; (iii) both, but mostly father; (iv) both, but mostly mother; (v) both, mother and father equally; (vi) other. We construct a multinomial variable, M_h, which takes values 0 in exclusively patrilineal villages (41.5% of villages), 0.25 in villages where patrilineal inheritance is dominant but not exclusive (26.8% of villages), 0.50 in villages where matrilineal and patrilineal inheritance is equally prevalent (7.3% of villages), 0.75 in villages where matrilineal inheritance is dominant but not exclusive (14.6% of villages) and 1 in exclusively matrilineal villages (9.8% of villages).

- *Returns to child-directed speech*: we ask parents their beliefs about the returns to child-directed speech under a high and low-investment scenario. Using 10 beans to reflect probabilities, we ask mothers and fathers to: "Pick the number of beans that reflects how likely you think it is that a child puts 2-3 words together in speaking by age 2 years of his/her life..."

  (a) if the carers frequently talk to the child directly?

  (b) if the carers rarely talk to the child directly?

  We subtract (b) from (a) to construct the measure of returns.

- *Cost of child-directed*: We ask parents the following question to measure costs: "On a scale from 0 to 10, where zero means not costly at all and 10 means very costly, how costly do you think it is to talk directly to your child?"

[47] We also check that our results are similar using the 5 seconds LENA threshold.

- *Vocalization*: for every 10 ms frame, the Voice Type Classifier (VTC) returns whether the key child (i.e., the child wearing the USB device), adult female, or adult male was vocalizing. A vocalization is defined as a sequence of frames in which a speaker vocalizes. We count the number of vocalizations attributed to the relevant speaker (i.e., key child, adult female, or adult male) over the whole recording. We then divide that measure by the length of the recording to control for variation in recording length. Our final measure is the number of vocalizations per day of recording.

- *Conversational turn (CTC)*: counts the number of inter-turn intervals between the speaker and the child using some simple assumptions. A vocalization is considered a turn if: 1) the vocalization is from the key child or the speaker (or a list of speakers), 2) the previous vocalization was from the other (if the vocalization is key child, the previous one has to be the speaker and vice versa), 3) the vocalization must have begun less than one second after the previous one ended (it can also begin before the previous one ended i.e. the new speaker 'interrupts' the previous one).

- *Pitch*: for each each adult speaker (i.e., adult female, adult male), we first measure the median pitch of each vocalization and then calculate the mean of the median pitch across all vocalizations in the entire recording. We calculate the mean of the median pitch under two conditions: (i) when the speaker is vocalizing during a conversational turn, (ii) when a speaker is vocalizing in the absence of a conversational turn. In our analysis, we use *pitch in the presence of conversational turn* to proxy for quality of child-directed speech. Pitch is measured in semitones.

- *Vocs at High Pitch*: for each adult speaker (i.e., adult female, adult male), we count the number of vocalizations for which the median pitch is higher than average pitch during a CTC.

E Survey Instrument

E.1 Sample recruitment

Through a public village lottery, we randomly selected 820 adults (women and men above the age of 18), 630 of whom have children, and roughly half of whom were married to one another and parents of a small child (6 months to 4 years old). The subset of 182 married couples with a small child, for whom we surveyed both the wife and the husband, constitutes our analytical sample. In a few cases in which the parents had more than one child in the target age range, we collected information on the additional children, for a total of 196 children.

E.2 Survey data

We conducted a household survey with mothers and fathers, individually. We collected basic demographic information including age, education, economic and productive activities, time use, asset ownership, social preferences, and risk-sharing arrangements, including financial, emotional, and childcare support. We asked mothers, only, to report information on their

children, including age, gender, health history, and school enrollment, and we measured the BMI of target children using standard practices recommended by the World Health Organization (WHO).

In addition, we collected experimental measures of cooperation through an incentivized dictator game in which each subject decided how to allocate an endowment of SI$40 ($5 USD at the time of our study, which is equivalent to the average daily GDP per capita) between herself/himself and a series of receivers. Receivers included: a random person in a distant village, a random person in the participant's own village (hereafter, "strangers"), and the subject's: spouse, mother, mother-in-law, father, father-in-law, sister, brother, neighbor, and close friend. We examine the share sent to the spouse as a measure of intra-household cooperation.

We also conducted a competition game in which subjects completed an incentivized task (i.e., the completion of visual puzzles, similar to Raven's Tests) for which they could decide to be paid a piecemeal rate or compete against an opponent (where they are paid for the correct number of completed puzzles only if they beat the opponent). Parents were offered the option to compete against their spouses and we document whether they chose to compete as a measure of intra-household discord. We also use the number of correct puzzles (out of 24) as a measure of cognitive ability.

To capture village-level information, we conducted surveys with the village leaders. We collected information on the inheritance practices of the village with the following question: "How is land inherited in this village?" - (i) from father; (ii) from mother; (iii) from both, but mostly father; (iv) from both, but mostly mother; (v) from both, mother and father equally; (vi) other. We also asked about post-marital residence practices to determine the degree of matrilocality (i.e., whether the groom lives with the bride's family upon marriage) using the following question: "When people in this village marry, does the couple live in the bride's village or in the groom's village?" - (i) bride; (ii) groom; (iii) it depends, but mostly bride; and (iv) it depends, but mostly groom. Finally, we asked about the predominant languages spoken in the village, the existence of public goods such as kindergarten, primary school, and secondary schools, whether there was a market in the village, and the type of leader that governs the village (i.e., elected, traditional, religious, elders, or a village committee).

E.3 Measurement of Beliefs

Parents were read the text below and asked to respond to the corresponding questions.

Now I am going to ask you some questions about your beliefs regarding certain behaviours that adults in your community could have and its effect on children. Before that, let's talk about how I am going to understand your answers better.

I will ask you several questions about the chance or likelihood that certain events are going to happen. There are 10 beans in the cup. I would like you to choose some beans out of these 10 beans and put them in the plate to express what you think the likelihood or chance is of a specific event happening. 1 bean represents 1 chance out of 10. If you do not put any beans in the plate, it means you are sure that the event will NOT happen. As you add beans, it means that you think the likelihood that the event happens increases. For example, if you put 1 or 2 beans, it means you think the event is not likely to happen, but

it is still possible. If you pick 5 beans, it means that it is just as likely it happens as it does not happen (50-50, like flipping a coin). If you pick 6 beans, it means the event is slightly more likely to happen than not to happen. If you put 10 beans in the plate, it means you are SURE the event will happen. There is no right or wrong answer, we just want to know what you think.

Let me ask you a couple of questions to make sure you understand how to answer using the beans.

Violations of monotinicity

1. Pick the number of beans that reflects how likely you think it is that....

   (a) A woman in your village will go to the market at least once within the next 2 days?

   (b) A woman in your village will go to the market at least once within the next 2 weeks?

Returns to speech

We now ask you to think about a hypothetical child born in your village this year and what may happen to this child depending on his/her carers' behavior.

2. Pick the number of beans that reflects how likely you think it is that a child puts 2-3 words together in speaking by age 2 years of his/her life...

   (a) if the carers frequently talk to the child directly?

   (b) if the carers rarely talk to the child directly?

Now, I would like you to think about your child (*ENUMERATOR: be sure to reference the child we are recording*) and how s/he compares to other children in your community.

3. Pick the number of beans that reflects how likely you think it is that...

   (a) child will be ready for school at age 6

      – if the carers frequently talk to the child directly?
      – if the carers rarely talk to the child directly?

   (b) child will learn well at primary school

      – if the carers frequently talk to the child directly?
      – if the carers rarely talk to the child directly?

   (c) child will go to secondary school

      – if the carers frequently talk to the child directly?
      – if the carers rarely talk to the child directly?

Costs of speech

We would like you to think about all the constraints regarding talking directly to your child. You may be too tired to do it or you may have no time to do it. All those things make being talkative costly. On the contrary, you may find that talking directly to your child is really easy and fun, and you easily find the time to do it.

4. On a scale from 0 to 10, where zero means not costly at all and 10 means very costly, how costly do you think it is to talk directly to your child?